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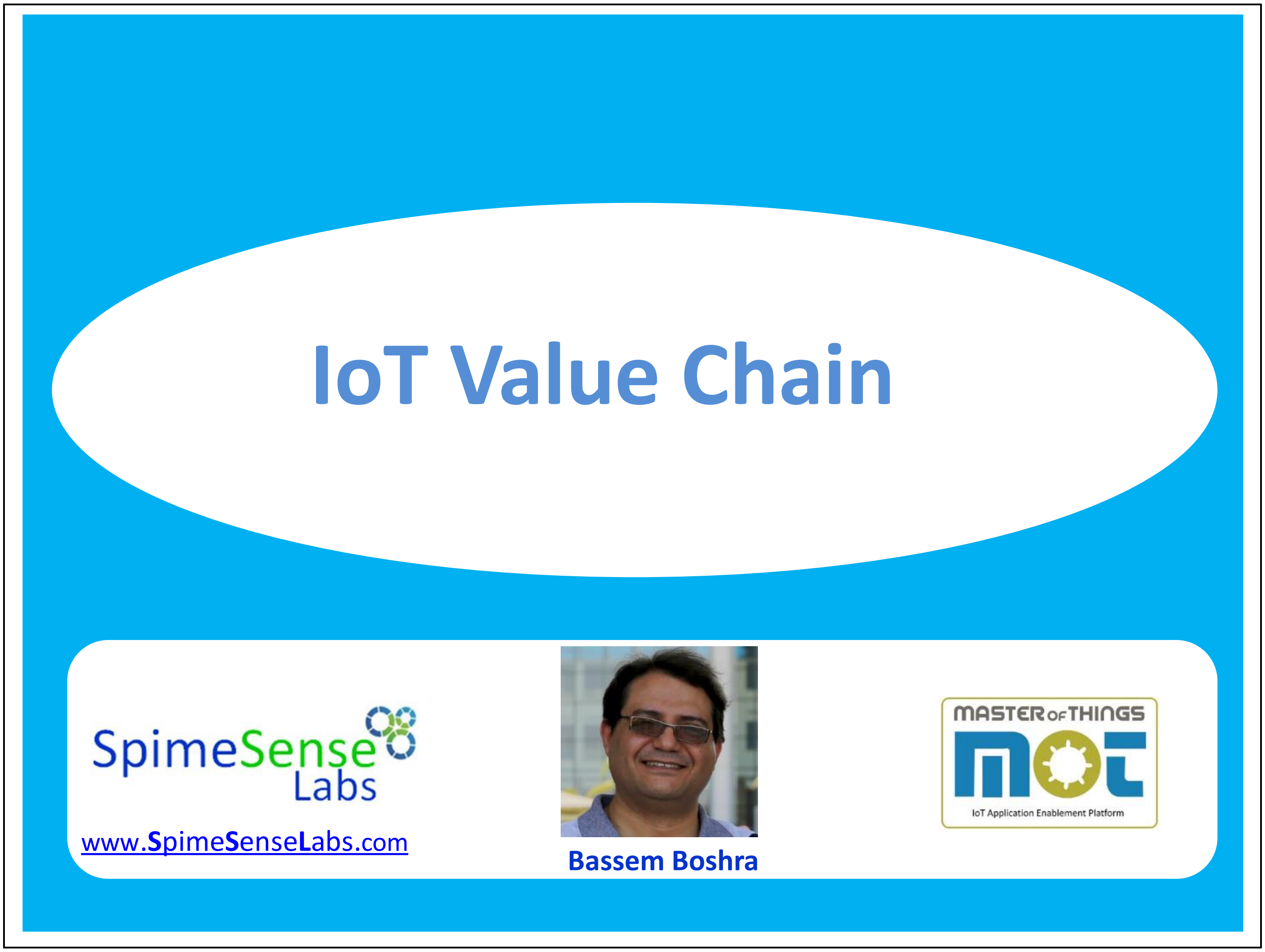


Eng. Bassem Boshra, the author of this material is a graduate of electronics engineering with post graduate studies from iTi (Information Technology Institute), Intake 8 in SSDP (software skills development program) and post graduate studies in Executive management from AUC. I worked as web applications instructor in iTi for 2 years, worked in NCR Internet Banking for 2 years, worked for Ericsson for 14 years in various technical, presales and executive management roles in Mobile Internet, mobile network management, real-time charging and billing adaptations (custom software) for telecom operators, I'm the 22nd Ericsson certified senior solution architect world-wide, used to be internal SI assessor to certify Ericsson architects and Project Manager worldwide and worked as an overall chief solution architect for telecom operators.

In 2013, I've founded SpimeSenseLabs with my friend and colleague Magdy Raouf and since that I'm deeply involved in IoT AEP, IoT solutions and IoT Networks for more than 10 years. I'm proudly being the consultant for my beloved iTi in IoT and Telecom.

About SpimeSenseLabs

About SpimeSenseLabs

Who are we?

SpimeSense Labs

SpimeSenseLabs (S.A.E) is an IoT company established in 2013 by expertise in software development, telecom & IT industry and relies on highly qualified young professionals.

Our expertise has more than 7 years of IoT experience and 25 years ICT industry, certified by global telecom vendors and have delivered innovative solutions to global telecom operators and banking institutions in the Middle East.

Our staff comprises highly qualified young professionals who have the merits of innovation among their basic skill set.

Our vision

Internet will be containing billions of connected devices exchanging data to serve the needs of human being turning the internet to be **IoT (Internet of Things) with great impact on our life and business** where we innovate platforms and applications to facilitate various industry verticals.

Dec 2019

IDC

Existing Customers
Active Trials

May 2017

Best Technology Enterprise

April 2016

Round 9 finalist award
MIT ENTERPRISE FORUM ARAB STARTUP COMPETITION

March 2016

Member of m2m alliance

Oct 2015

sigfox
Platform of the month

Nov 2014

Innovation Excellence Award
CAIRO INNOVATES
1st Cairo International Exhibition of Innovation
معرض القاهرة الدولي الاول للابتكار

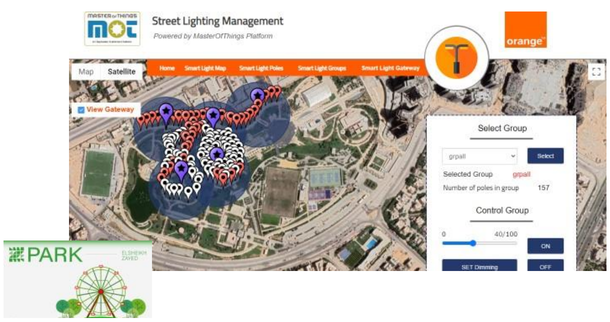
Where you can find MasterOfThings IoT AEP

MasterOfThings & SpimeSenseLabs

Became IoT Platform provider for all Orange Egypt corporate customers.

Selected by MCIT as platform for 10 IoT labs among Egyptian governorates.

https://www.linkedin.com/pulse/building-university-iot-labs-professional-development-bassem-boshra/



Recognized globally as a "Horizontal Platform" with "Broad offering" in 2021 IoT Application Enablement Platform (AEP) Scorecard report by MachNation

Report Source:
https://www.machnation.com/2021/02/21/machnations-2021-iot-application-enablement-platform-aep-scorecard-evaluates-13-internet-of-things-platform-vendors/?utm_source=newsletter&utm_medium=email&utm_campaign=machnation_publishes_2021_iot_application_enablement_scorecard&utm_term=2021-02-21

Delivered Smart Stations solutions for Upper Egypt

More info:
https://www.youtube.com/watch?v=Ddbps1GajcA
https://www.youtube.com/watch?v=gEVuCaXJg80

The Internet as it was known back in 2013 and onward till 2024

Some of the internet services as it used to be till 2013 where lots of services are offered to end users who were only using their computers or phones to access the service.

In late 2013, predictions for users to have devices/things connected to the internet other than computers and phones and the term of Internet of Things started to appear.

The term was originally initiated inside the telecom industry and it was called m2m as an abbreviation for Machine-to-Machine communication.

Later it was named as Internet of Things by Telecom and IT experts. In an interim stage Cisco came by another marketing name calling IoT as an abbreviation for Internet of Everything where Cisco pushed the name of fog computing as well.

It ended up to be called IoT as an abbreviation for Internet of Things.

When it comes to using IoT in industrial automation, it is referred to as IIoT (Industrial Internet of Things)

Still some people are trying to put other names for IoT use in some certain fields like IoMT (Internet of Military Things), IoFT (Internet Of Flying Things) and many more.

In 2021, Academia introduced a new term AIoT as an abbreviation for Artificial Internet of Things as the mix of 2 technologies Artificial Intelligence and Internet of Things but soon it faded out.

Today there are 2 common terms that are being used namely IoT and IIoT.

Internet evolution from internet of PCs to internet of Things

The introduction of internet connection in any device results in many new applications that utilize the device connection capabilities.

When end users had only the PC/Laptop connected to the internet, many web applications appeared (i.e. Facebook, Gmail, and YouTube, etc.) and with the introduction of Smart Phones getting connected to the internet resulted in many other applications appeared (i.e. WhatsApp, Viber and huge list of apps on Android Market), the introduction of a new connected wearable device (i.e a smart watch) led to many new applications (Activity trackers, Calory burning calculator, healthy body , your calendar events versus weather forecast, etc.).

Similarly, getting other devices connected to the internet will lead to new wave of applications making use of this device connectivity for the benefit of the users. Your car manufacture will offer you a applications on your car, on your mobile and web based applications to help you to utilized your connected car (i.e. monitor your car performance, self-check for car components, plan the car spare parts, etc.).

In the same way, when home appliances and meters get connected to the internet, new applications will appear to offer the end users extra benefits of these connected devices that was not possible before (i.e. monitor your house utility consumption, switch between postpaid to prepaid metering, analyze your utility consumption over the day, etc.).

While getting the land and plant connected leads to water savings and higher quality agricultural products (i.e. more tasty fruits, longer shelf live fruits, more healthy animals in the farm, etc.).

Getting all these things connected to the internet turns the internet to be Internet of Things, abbreviated as IoT, and that leads to appearance of lots of new applications every day to serve the owners of these connected things.

Comparing IoT forecasts between 2013 and its updates in 2025

2020 Estimates & forecasts versus actuals

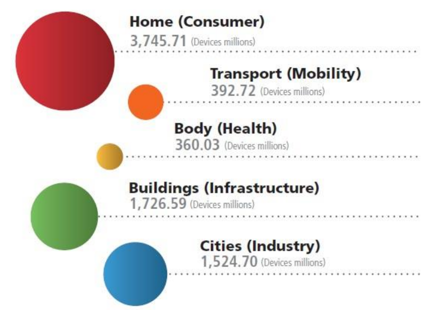
- Ericsson estimated 50 billion connected devices by 2020.
- Cisco estimated 50+ billion connected devices by 2020.

28
Billion
By 2025

- Harbor research estimates 8 billion devices(Excluding Mobiles, Tablets & Laptops)
- Harbor research estimates business value of 1 Trillion $

1.25$
End of 2018

  - **Managed services 571 billion $**
    - Systems development tools, Data & analytics
    - Value Added Application Services
  - **Enablement HW 53 billion $**
    - Connectivity modules (wireless & wireline)
  - **Network Services 387 billion $**
    - Wireless & Wireline operators business



- Pike research estimates 108 billion $ in Smart Cities technologies.

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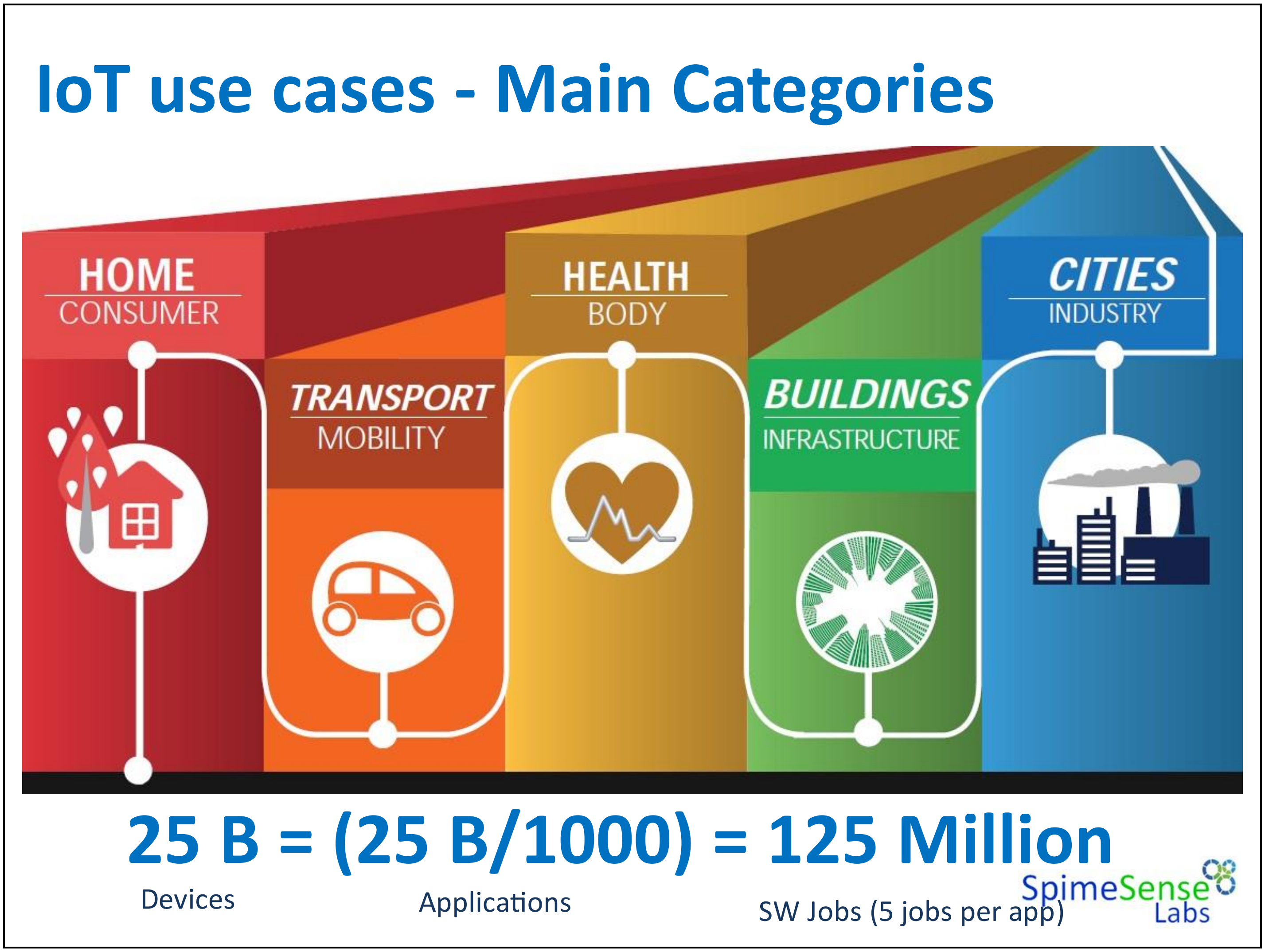
Looking at the historical estimates and predictions about the IoT and comparing it with what actually happened gives us the ability to judge the new estimates for the future and make these forecasts more realistic.

We can notice that the estimated number of connected devices in 2020 was much higher than what actually happened while the estimated amount of money in deals related to IoT was much less that what was estimated.

Please notice that most of generated money, more than 50%, was in software and applications while more than 30% was in connectivity and network services and around 5% was in enablement HW (modems) use to get the devices connected.

It is also important to note that around 10% of the whole money generated was in smart cities use cases including devices used in managing the cities, getting these network connectivity services for the city devices and software and applications used in operating and managing the cities.

The IoT 5 main pillars – categories of IoT use cases.



IoT use cases are categorized into 5 main categories and usually called the 5 main pillars. Each of these categories has its own characteristics.

Home & consumer pillar is characterized by the consumer perception for the high value of the device HW without being able to evaluate its software component at all.

Both Cities and Industrial IoT use cases are characterized by a perceived high value of the software. The city operation management users and manufacturing facilities management team realize how much cost savings and time saving can be archived in city operations because of the software that manages the city operations. Similarly manufacturing facility owners realize how much they can increase the productivity and make more money because of the software that monitors and controls the machines in the manufacturing facility.

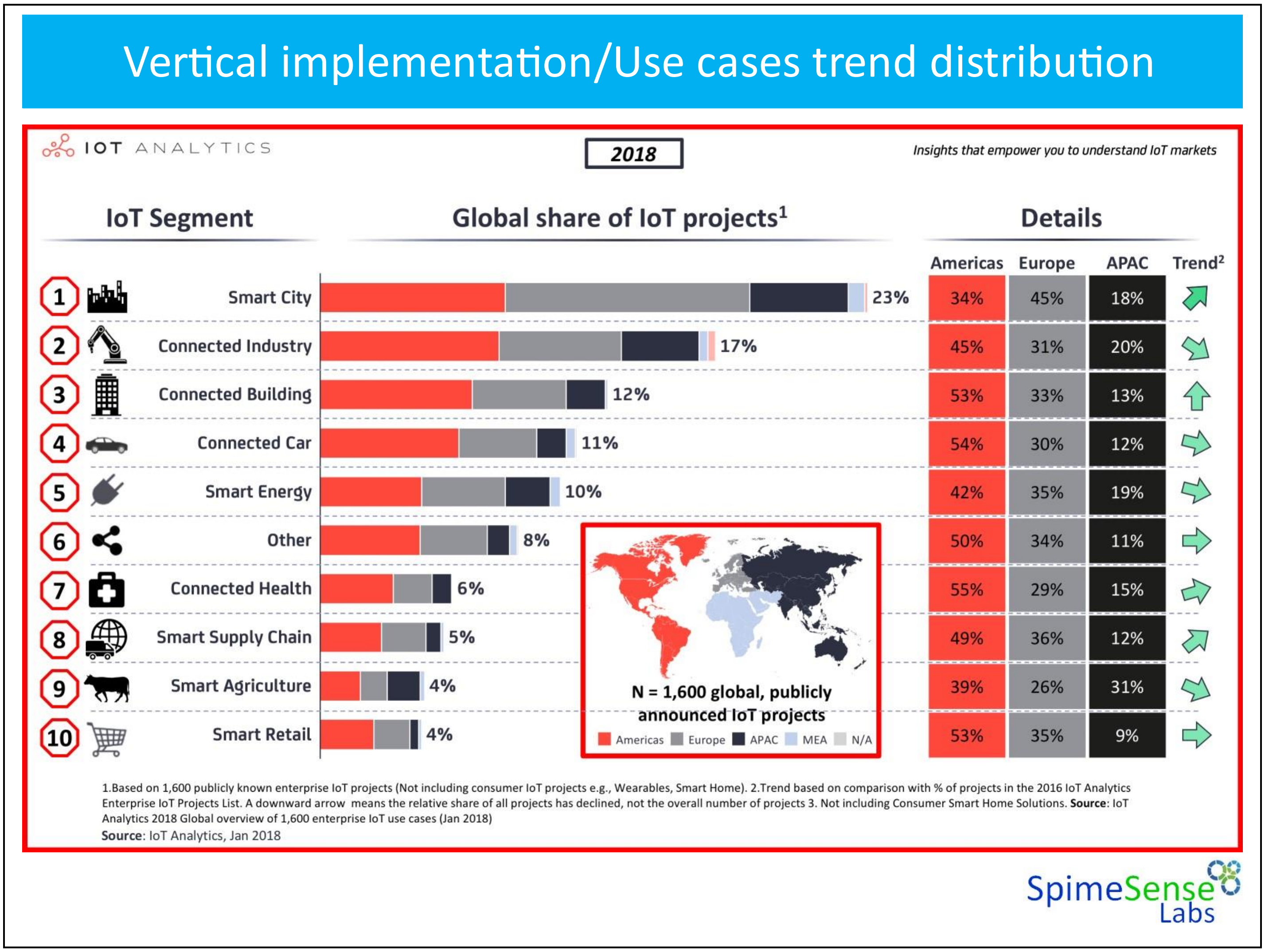
Health and body management use cases are characterized by high investment cost (in hundreds and thousands of million dollars) over a long period of time that can reach somewhere between 12 to 20 years of R&D and multiple phases of testing to reach medically approved IoT devices to be implanted inside the human body. it is also characterized by high ROI as well.

Transport and mobility use cases are characterized by huge data traffic generated over the networks due to connected transport and mobility devices.

Buildings and infrastructure are characterized by technology lag and always being behind due to relying on accumulated past experience of consulting companies who plan for these projects. It is totally depending on long accumulated past experience of the consulting companies who would prefer to use only guaranteed products with proven historical record in the filed to keep their image in front of their clients and hence most of the building and infrastructure projects will be utilizing technology from the past that proved itself in previous projects over many years.

There are other categories of IoT use cases (i.e. gaming and military use cases) but those will be much less than the 5 main pillars.

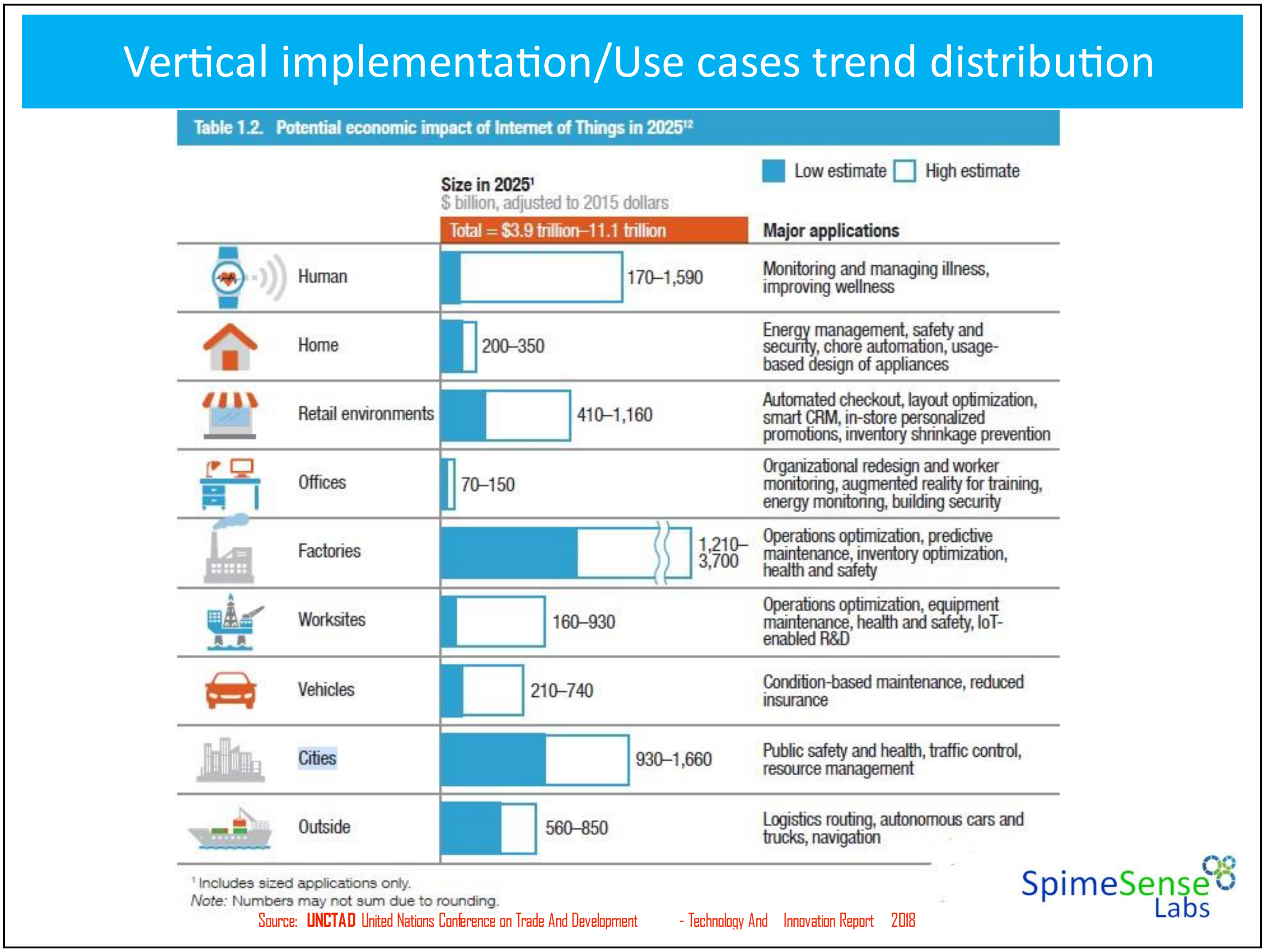
Categories of actual 1600 IoT projects implemented in 2018



Studying the actual implemented projects or announced projects with approved budget during 2018 shows that the biggest number of projects was in Smart Cities and 2nd to it is the number of projects in connected industries (IIoT/Industrial IoT).

It worth noting that the minimum number of projects were in Smart Agriculture and smart retail project.

World Wide planned budgets for IoT projects until 2025



	Size in 2025[1] $ billion, adjusted to 2015 dollars Total = $3.9 trillion–11.1 trillion	Major applications
Human	170–1,590	Monitoring and managing illness, improving wellness
Home	200–350	Energy management, safety and security, chore automation, usage-based design of appliances
Retail environments	410–1,160	Automated checkout, layout optimization, smart CRM, in-store personalized promotions, inventory shrinkage prevention
Offices	70–150	Organizational redesign and worker monitoring, augmented reality for training, energy monitoring, building security
Factories	1,210–3,700	Operations optimization, predictive maintenance, inventory optimization, health and safety
Worksites	160–930	Operations optimization, equipment maintenance, health and safety, IoT-enabled R&D
Vehicles	210–740	Condition-based maintenance, reduced insurance
Cities	930–1,660	Public safety and health, traffic control, resource management
Outside	560–850	Logistics routing, autonomous cars and trucks, navigation

[1] Includes sized applications only.
Note: Numbers may not sum due to rounding.
Source: UNCTAD United Nations Conference on Trade And Development - Technology And Innovation Report 2018

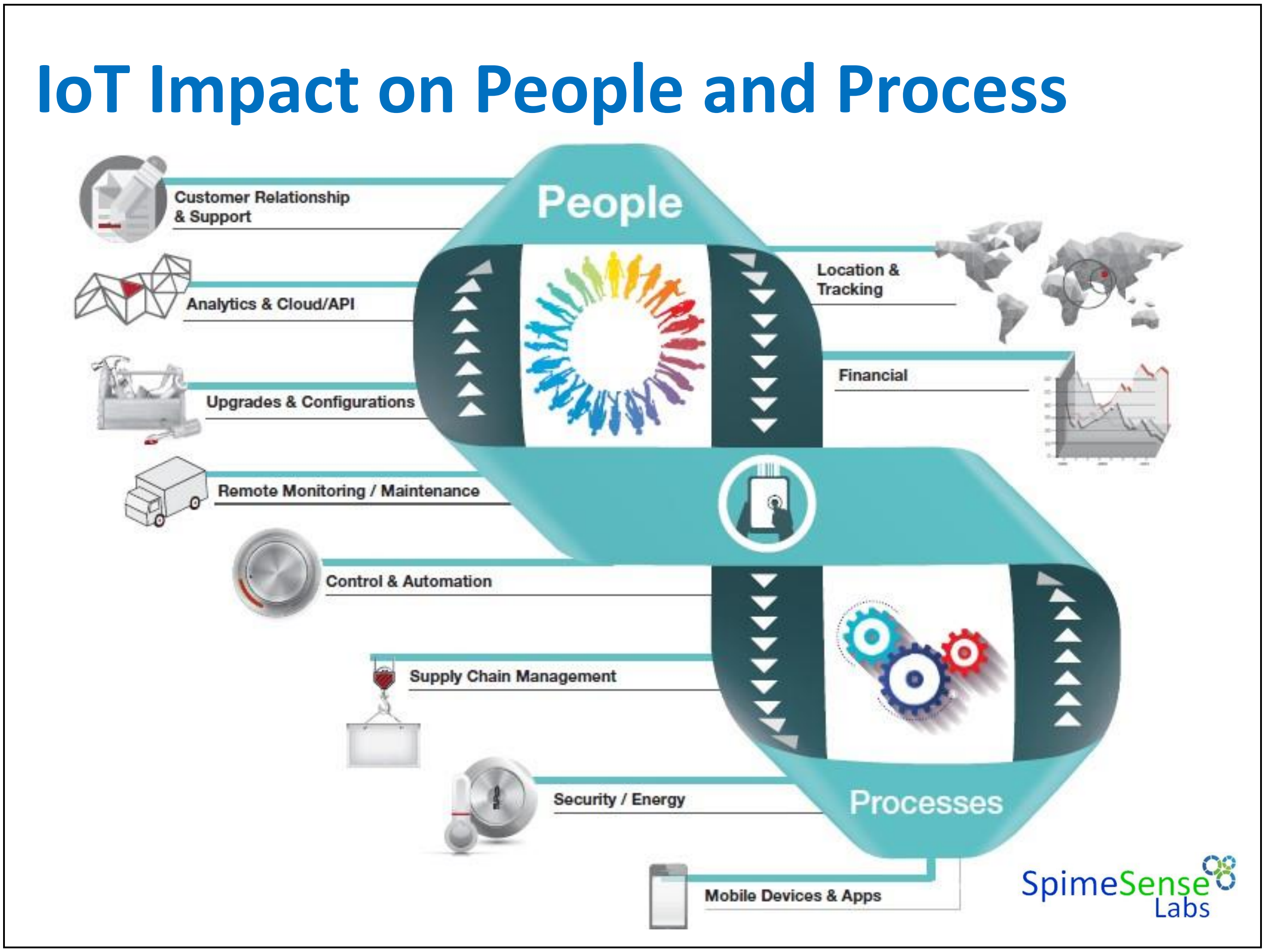
UNCTAD summary for planned IoT projects for 2025, according to inputs of the attending members, show that the biggest budgets are planned for connected factories the 2nd to it is the smart cities.

The smallest IoT projects during 2025 are in the office management, construction sites management and home automation & management.

It is worth noting that the tope 2 categories in 2025 are still the same as the top 2 back in 2018. Even that both industries have exchanged the position of 1st and 2nd but still taking the position on top of all other categories.

It also worth noting the rise in shipment and logistics in geographical areas not covered by terrestrial/on-land connectivity networks and hence the rise of satellite IoT connectivity.

IoT solutions impact on people and process in various businesses



IoT has a huge impact on people and process in all business domains to save time, to save cost, to have a better ROI and higher profitability.

One of the most impacted domains is the customer relationship management. This domain will witness a reverse of the whole process. Before IoT customers used to call short codes to get access to customer services through IVR (Interactive Voice Response) while with IoT, customers will be proactively notified about their services and proactively contacted by their service providers.

Having your cars connected will lead to having your "Car service center" automatically calling you to notify you that you have an issue in your car that requires your attention and notify you've got a booked time slot to handle this issue for your car in 2 days from now.

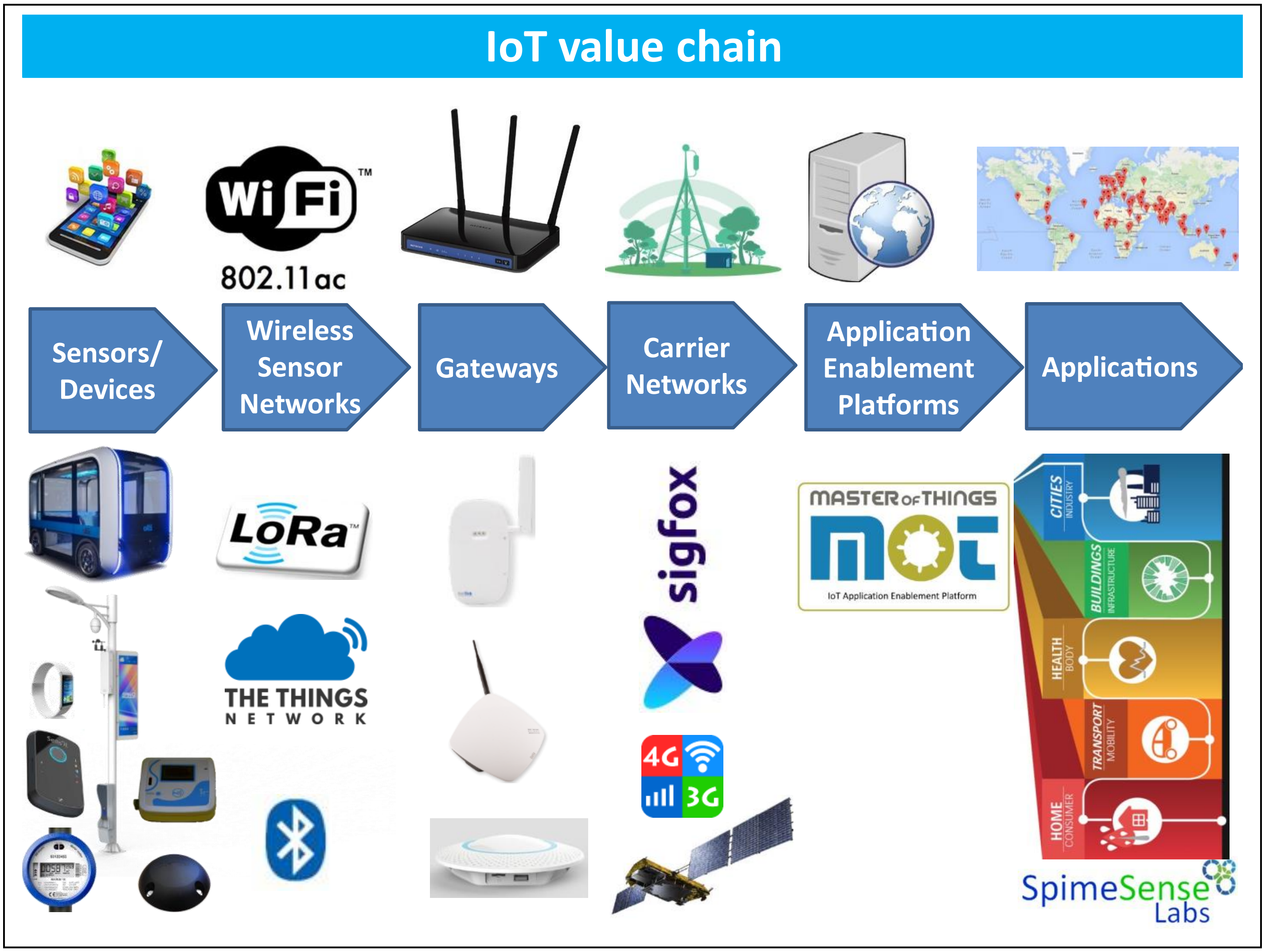
Similarly having your home appliances connected will lead to offering proactive maintenance and support by an authorized service center in your area calling you to propose a time slot to visit you to repair your home appliance (washing machine, air condition, etc.) that is functioning inefficiently.

Supply chain is significantly enhanced where most of the components planning (i.e. required spare parts) are based on actual needs calculated based on data collected from the field (i.e cars running in the streets or home appliances operating in your home) compared to best estimate planning based on previous experience.

It is worth noting that the introduction of software component in all of the things around us will result in higher level of satisfaction where consumers will get regular enhancements and sometimes new features on their devices (i.e. home appliances, connected cars, etc) due to remote FOTA (firmware over the air) updates and if the firmware of these devices were not developed properly with the right tools that cater for security and privacy during the development process, it will reach the end user devices with much less privacy and more insecurity where all of the end user devices will be remotely monitored and contacted by its manufactures and suppliers with good intentions from the suppliers to offer a better service to the end user but still have open door and possibilities for bad intention remote access both others.

It is important for software developers to think how these remote device monitoring and remote device FOTA update will happen? This will be clarified during the IoT applications development course through the use of MQTT protocol.

The IoT value chain, the building blocks of IoT solutions



Understating the building blocks of internet of things and the value of each block.

It is easier to understand the IoT solutions value chain (building blocks) by comparing it to the mobile solutions value chain. The provided figure illustrates the path of the mobile application data traffic in the upper part and compares it to the IoT data traffic path in the lower part.

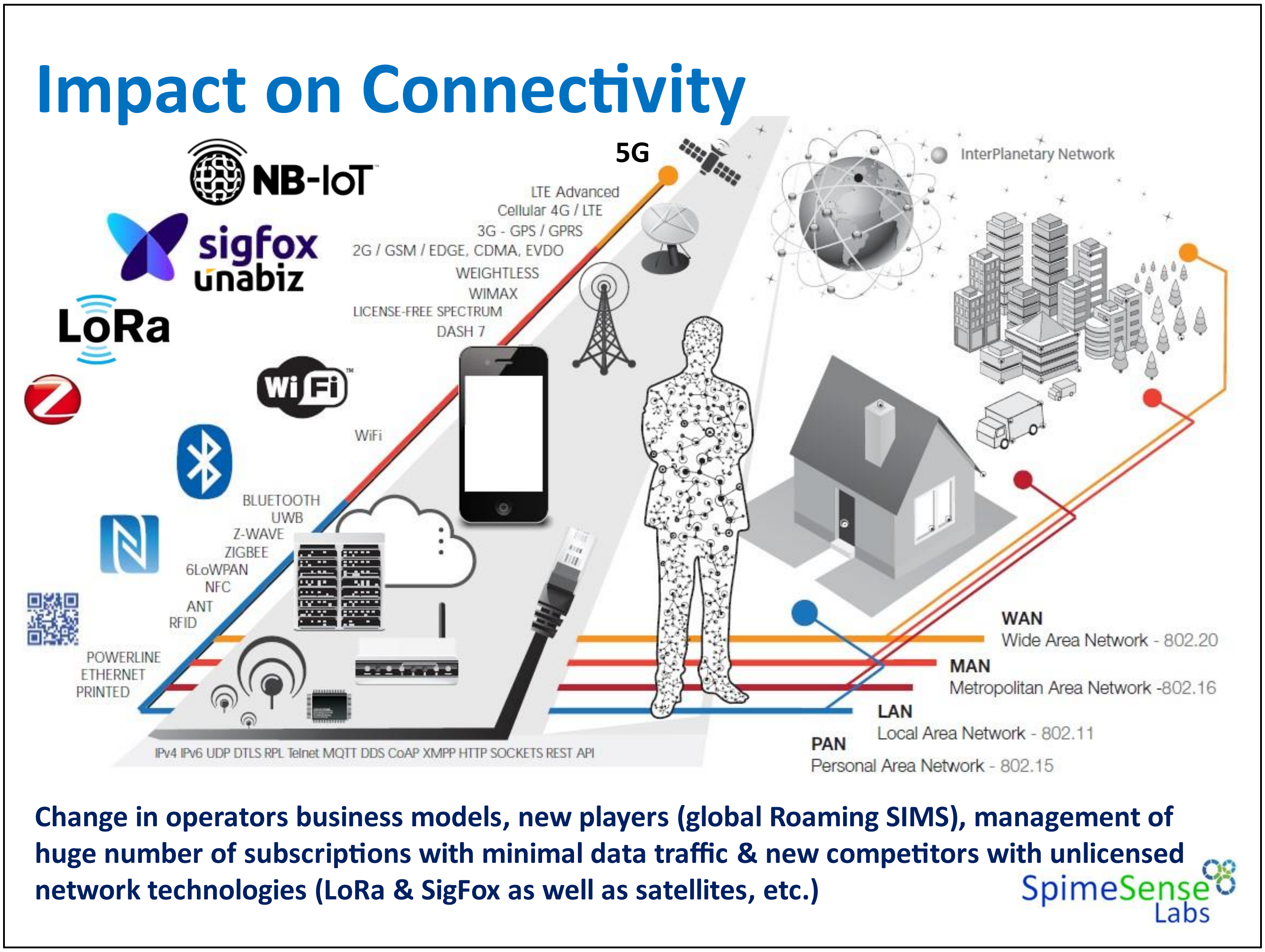
In the mobile world, the building blocks start by the mobile phone (the end user device) and it runs an application client (i.e. WhatsApp, Facebook, etc.) on top of the phone operating system (i.e. Android, iOS, etc.). When the mobile is being used at home, the mobile sends the data through its local Wi-Fi network to the Wi-Fi access point/router at home (the Wi-Fi Gateway) and the router transfers the data to the internet through a carrier network (ADSL copper cables, fiber cables, 3G/4G/5G network or satellite network) that carry the traffic though a very long distance (sometimes hundreds of kilometers) to its final destination application server on the internet (cloud server on the internet) .

Similarly, the IoT components and building blocks start by IoT Sensors/Devices (computerized device similar to mobile phone) whether it is a smart meter, street lighting pool, digital signage screen or parking sensor.

The sensor/device data is transmitted through a small distance wireless network called wireless sensor networks (somehow similar to Wi-Fi network at home) to reach the IoT Gateway (very similar to Wi-Fi Access Point). The IoT Gateway transfers the data to one of the carrier networks that carry the data through a long distance to its final destination on the Application Enablement Platform (compared to application servers) that runs a server-side application (compared to web application).

Each of those components and building blocks is explained in more details in later sections.

Characteristics of various IoT wireless technologies



The provided picture shows a curve going from the bottom left corner up to the upper right corner with all the wireless communication technologies presented on the curve. The more we go up in the curve, the longer the distance data is transmitted or the higher the data rates transferred using a wireless technology presented on the curve.

The curve starts on the left bottom side by the QR-code (similar to bar codes) printed on the products. We see these QR codes and bar codes printed on all the products in the super market and it includes data about the product and is transmitted to cashier local computer using infrared. If the local computer cashier will transfer the data later to the internet, it is important to note that the data included in the QR-code is fixed data and never change and remains as it is printed on the product package even if the product remains on the supermarket shelf or store for few years and also notice that there is no power source is available in the product package.

Similarly, if we go up in the curve, we will notice the introduction of NFC (Near Field Communication) and RFID (Radio Frequency Identifier). Both are wireless technology used in tags (stickers) or plastic identification cards. Each tags holds information about product and sticked/glued on the product replacing the printed QR-Code on the product. Information stored on the RFID Tags or NFC cards remain the same. and still there is no power source provided with NFC cards, NFC tags, passive RFID tags and all the power comes from the reader (NFC reader or RFID reader) that emits power to the tag/card to wirelessly charge the tag with enough power for the tag to be able to transmit its stored information back to reader.

RFID and NFC are both short distance, low power and very limited amount of data or even static data.

Radio-Frequency Identification (RFID) and Near Field Communication (NFC) are both wireless communication technologies, but they differ in several key aspects.

While RFID and NFC share similarities in their wireless communication capabilities, they differ in terms of communication range, frequency, speed, modes of operation, application focus, security features, and integration with devices. The choice between RFID and NFC depends on the specific requirements of the intended application, considering factors such as range, data transfer speed, and security considerations

Going further up in the curve we can will find more wireless technologies that are capable of either transferring more variable amount of data back and forth with either more data throughput or to a longer distance between the transmitter and the receiver. The transmitter is usually a modem installed inside the IoT device that transmits device data to the remote receiver installed in the IoT gateways like the case in BLE (up to 300 m in BLE 5) and the case in LoRa (up to 15 km depending on the antenna size). Receiver can be installed in telecom towers (in 2G BTS tower, in 4G eNB tower or in 5G GNodeB tower) or even installed in a satellite in the sky.

It is important to notice that going up in the curve these wireless technologies on the curve will enable us either to transmit more data with higher data rates or transmit the data to more longer distances as we go up in the curve. In both cases (higher data rates or longer distances) more power is consumed from the device. On the contrary, to save the IoT device power, we either use a wireless technology that sacrifice the data rate or sacrifice the distance.

Data can still be transmitted a to a very long distance (1,000 KM and more in case of satellites) with very low power and small antenna but that has to sacrifice the data rate and similarly, you can transmit data in higher data rates with very low power but that has to sacrifice the distance (i.e. NFC). **You can never have both higher data rates transmitted to long distance at the same time with low power consumption and a small antenna as well.**

a team of scientist **In Nov 2021**, the team consisted of Thomas Telkamp, CTO of Lacuna Space and Frank Zeppenfeldt, a 20-year ESA veteran working in the field of IoT and Smallsats, the team managed to transmit and receive a LoRa signal **amplified to 350w using a 25-metre dish antenna** of the Dwingeloo Radio Observatory in the Netherlands to the moon and received the signal reflected by the moon back to earth on the same antenna for a distance of 730,360km (double the distance between earth and moon) for the furthest distance such a data message has travelled.

Here is a nice and simple experiment you can do yourself using a standard android phone to understand how power consumption relates to distance. Download google maps offline on your local phone to use it offline. After that, wait for the night (away from the sun light) and fully charge your phone battery then turn off all the data traffic and WiFi. Then reduce the screen light/power to the minimum.

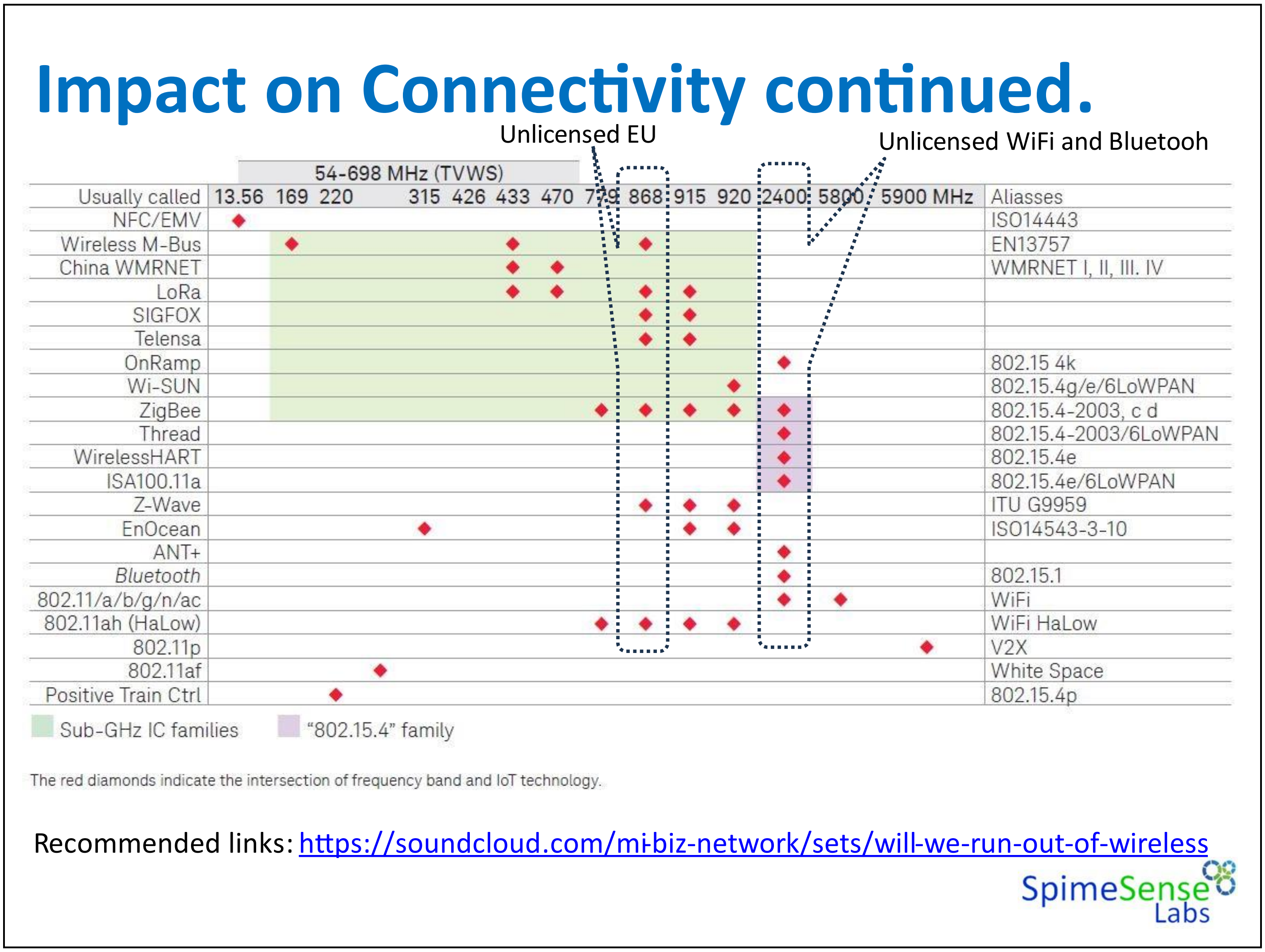
Now go out in the street will your phone battery fully charged (100%) and turn on your GPS and open your offline google maps. Keep moving around your house for half an hour. Your phone will keep sending and receiving data to and from the GPS satellite while you are moving around your home. You will notice that your phone had been able to recognize its location by regularly sending the phone GPS chip ID to the GPS Satellites (depending on your phone it might use other positioning satellites like GLONASS, BeiDou or Galileo) flying in MEO orbits 20,000 K meters above the earth and receiving the GPS satellite response back. After this half an hour, record the battery level of your phone and notice How much did your phone consume of the battery per minute and calculate the battery loss rate per minutes. For example, you have started your walking with 100% battery and kept walking for 10 minutes and then the battery went down to 95%, this means you lost 5% of the battery in 10 minutes. That is equivalent to O.5% per minute. Please post this information as a comment on this article. Tell us your phone brand, model and the percentage of phone battery per minutes. In this experiment we are assuming your phone battery is healthy and good.

Experts' advice on What is the best way to design an IOT network for a smart city, is available at https://www.linkedin.com/advice/1/what-best-way-design-iot-network-smart-city-ynb8c?utm_source=share&utm_medium=member_android&utm_campaign=share_via

More information on **What Has To Happen For 5G To Be Deployed In Smart Cities** is available at

https://www.linkedin.com/pulse/what-has-happen-5g-deployed-smart-cities-smartcityexpoworldcongress-dqscf/?utm_source=share&utm_medium=member_android&utm_campaign=share_via

Frequency bands use for wireless IoT technologies.

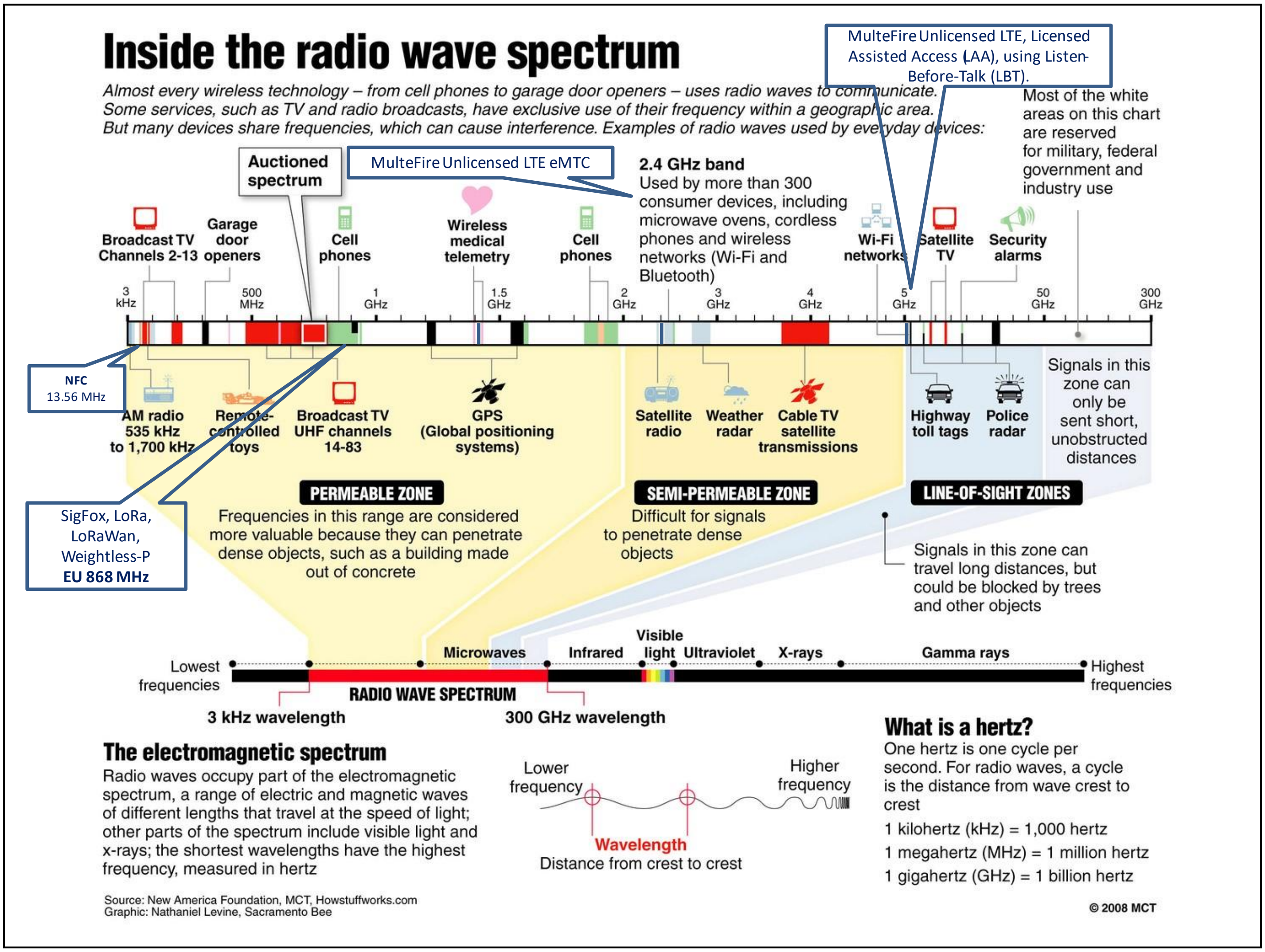


The above figure lists some of the wireless technologies used in wireless communication and the frequency band that can be used in each of these wireless technologies.

Please note the frequency bands highlighted by doted rectangle. The 868-frequency band is possible to be used by most low power wireless technologies for long distance communication. This band is sometimes referred to as ISM (Industrial Scientific and Medical) and that is the band used for unlicensed communication in EU countries and some of the northern African countries (i.e. Egypt). Similarly, 2.4G (2400) hertz is also unlicensed frequency band that is used by WiFi and Bluetooth wireless communication technologies with BLE (Blue Tooth Low Energy) used for saving the power on the IoT device battery.

Recommended links: https://soundcloud.com/mi-biz-network/sets/will-we-run-out-of-wireless

Closer look inside the radio wave spectrum allocation.



Any useful communication will require a range of radio frequencies. In a single word it requires “spectrum” and the more devices communicating, the more spectrum is needed.

The above chart is called the frequency spectrum map. That map shows spectrum versus the services allowed to use that radio frequency range. Each country regulates the spectrum map to allocate/dedicate frequencies to certain services to overcome any interference between various radio waves used for various services.

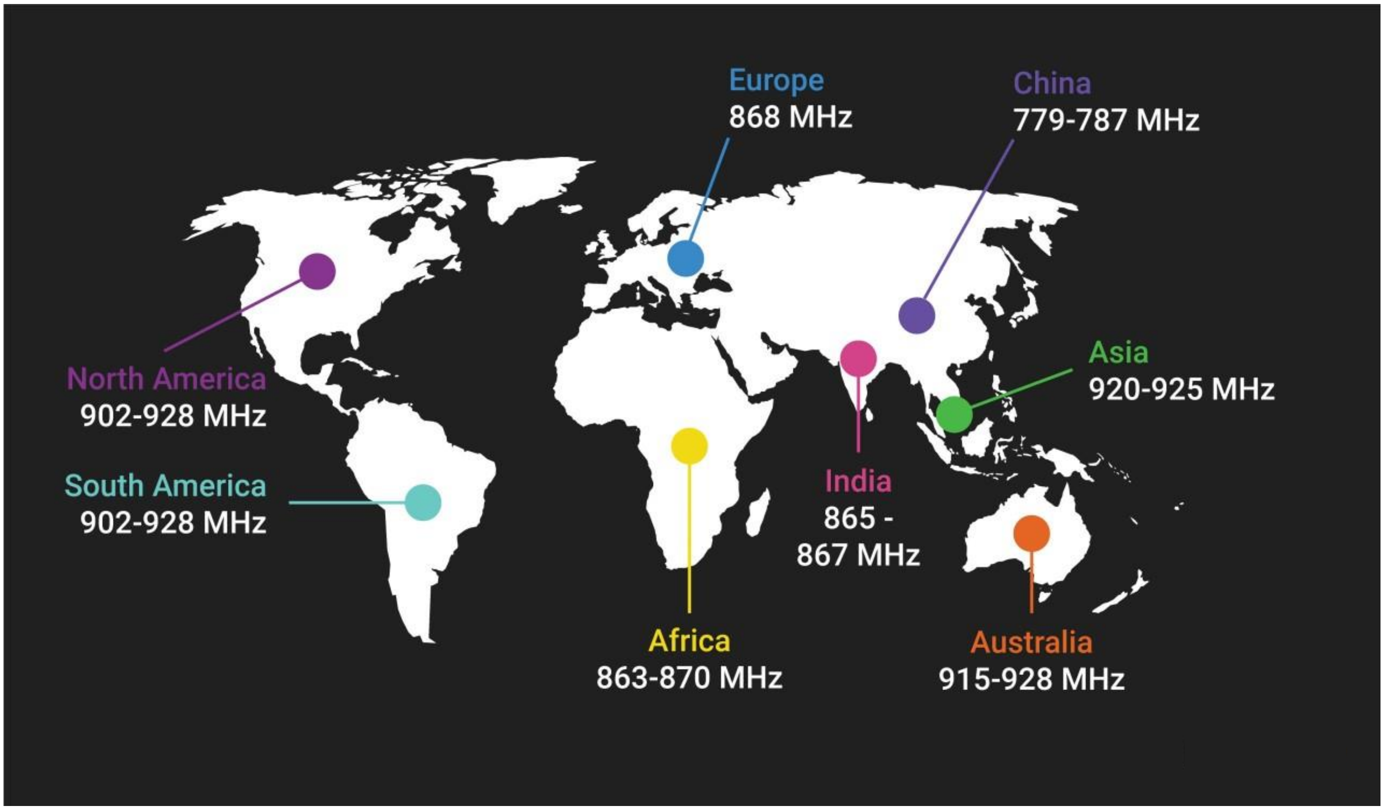
Part of the spectrum is allocated for television, part of the spectrum is allocated for special purposes emergency services, part of the spectrum is allocated to mobile communication services and part of the spectrum is allocated for military purposes.

Of particular interest is a range of frequencies around 868 MHz that has been freed up for ISM (Industrial Scientific and Medical) use and also being used by IoT wireless communication in European Unition countries and some African countries including Egypt.

It is important to note that the spectrum map would differ from geographical region to the other where group of countries agree on certain frequencies to be allocated/dedicated for certain services. All of the countries came to an agreement that certain frequencies are dedicated for certain services everywhere in the world (i.e WIFI, Bluetooth, GSM, 2G,3G, 4G, satellite TV and Positioning Systems) but unfortunately not all the world agree on one map for all services and hence the unlicensed IoT wireless communication range would differ from geographical region to the other and from a country to the other.

The unlicensed frequency range for IoT networks world wide

Unlicensed frequency range in every region



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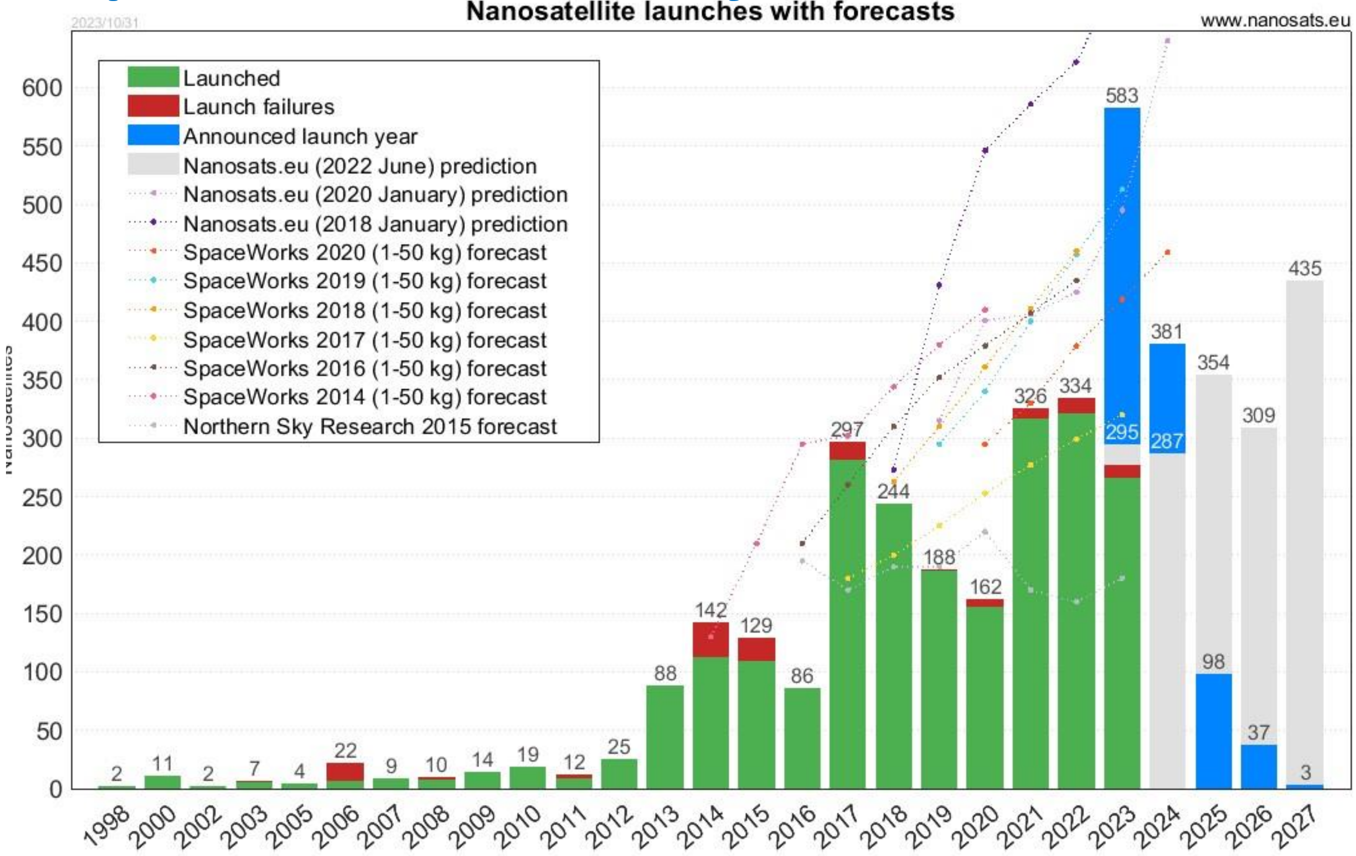
The map shows the frequency ranges available for IoT unlicensed wireless communication in various geographical regions in the world. It is important to notice that IoT devices that will work in EU countries **will work in Egypt** as well. While IoT devices that are made for other geographical regions **will not be accepted to operate in Egypt** as it would cause frequency interference with other services and that might lead to life threating consequences specially if this interference affects the medical devices in hospital ICU (intensive care units) or in military use.

Because radio waves travel for a long distance, you can imagine the disasters that could happen if an interference happens due to using wrong frequency. This interference could affect a flight control system in the sky or could affect a military mesial control system on the ground and causes the mesial to fire and hunt down a civilian flight with hundreds of travellers.

The rise of LEO cub satellites for wireless communication

Impact on Connectivity continued.

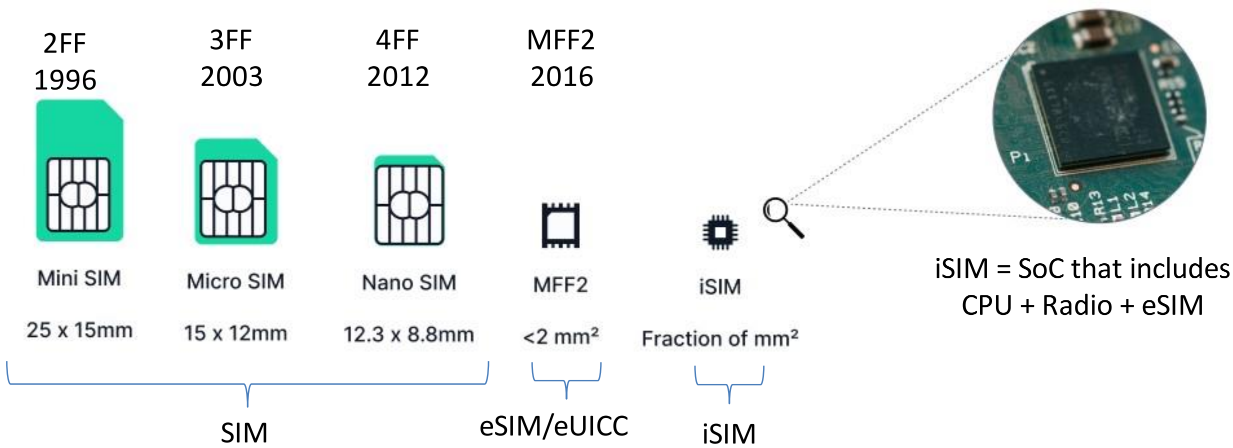


The chart shows the number of the LEO (Low Earth Orbit) satellites launched since 1998 and planned for launching in each of the coming few years. Summing the numbers in all years will end us of thousands of LEO satellites circulating around the earth. This will end us to have good services in the coming few years but on a long term increases the space debris.

Please note that these LEO satellites are circulating in an orbit of an altitude between 500Km to 1000km on top of the earth surface and circulating with a speed of 28,000 Kilometer per hour (depending on the altitude) to overcome the impact of the earth gravity however their life time would be around 5 to 8 years depending on their altitude before these satellites are grabbed by earth and burn due to friction with the air in the earth atmosphere.

From SIM to eSIM and iSIM for NB-IoT (mMTC) in 4G & 5G

Connectivity: SIM Form Factors for 4G/NB-IoT/5G



Embedded SIMs (eSIMs) or embedded Universal Integrated Circuit Cards (eUICCs) are physical SIMs that are soldered into the device and enable storage and remote management of multiple network operator profiles (remote SIM provisioning). The form factor of eSIM is known as MFF2.

The integrated SIMs (iSIMs) moves the SIM from a separate chip into a secure enclave alongside the application processor and cellular radio on a purpose-built system on a chip (SoC).

Good Tutorials : https://youtu.be/oWJiSaKLXnE
https://youtu.be/ZAe9GSpZdr4

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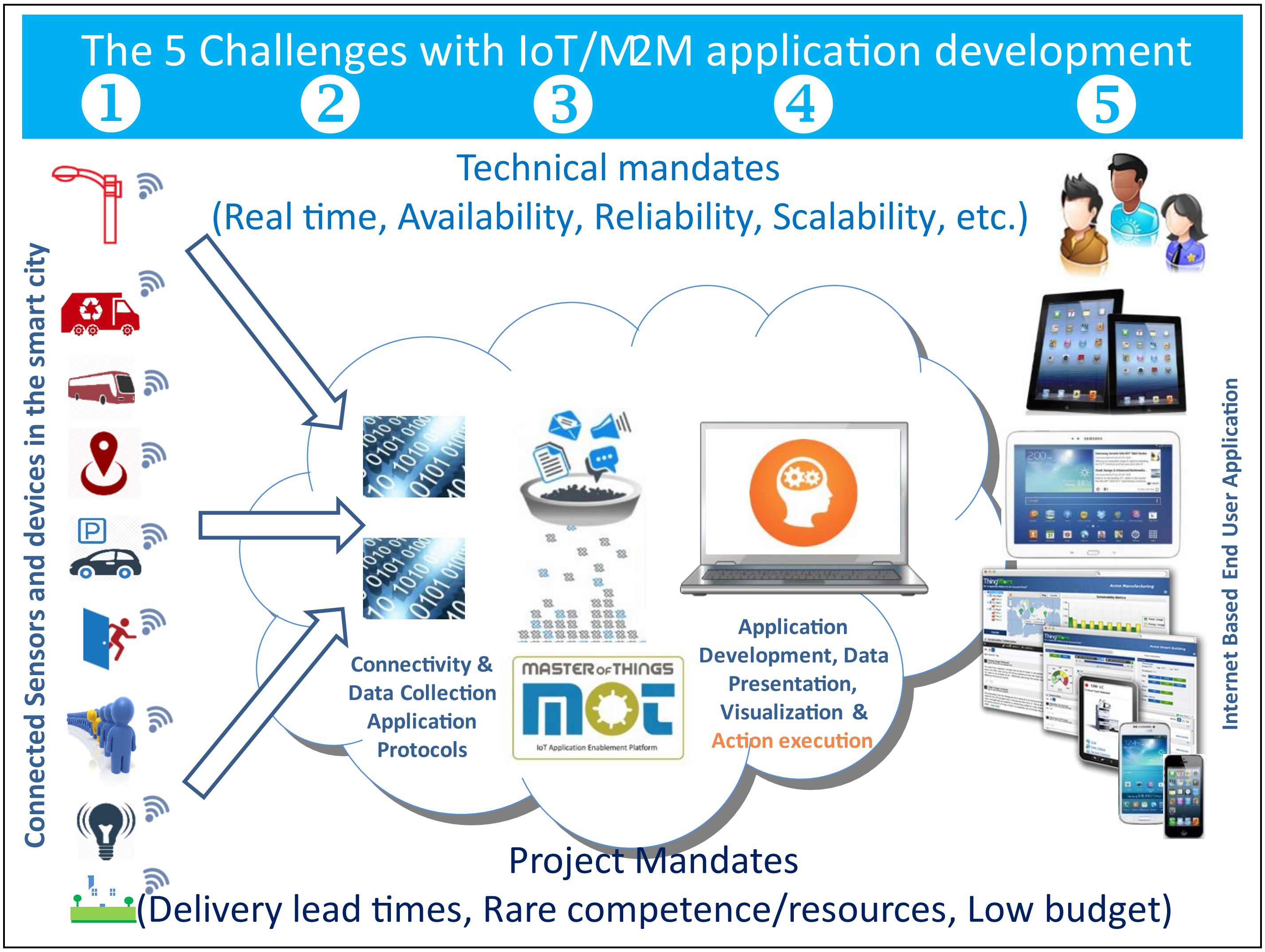
Various form factors of the SIM (Subscriber Identity Module) chips have been used for mobile communications over many years.

For IoT devices that will use NB-IoT for connectivity in 4G or 5G, these devices could use the option to have an eSIM or iSIM. eSIM (eUICC) allows the devices to change its SIM number without having to change a physical SIM.

iSIM has the same functionality as eSIM but also makes the functionality built in the radio communication chip as well as the CPU chip. This should make it cheaper when it comes to the overall device cost.

Good tutorials about eSIM and iSIM are available in the YouTube videos in the links https://youtu.be/oWJiSaKLXnE and https://youtu.be/ZAe9GSpZdr4

The challenges, technical & project mandates in IoT solutions.



Building IoT solutions requires a solution for 5 IoT challenges and satisfying Technical and Project Management mandates explained below.

1st challenge is the domain knowledge. Nobody can solve a problem before understanding the details of the problem itself. If you plan to build an IoT solution for preventive maintenance for certain home appliance (washing machines), then you need to understand how the washing machine works and what are the main reasons for faults and hence maintenance requests in washing machines. It is the responsibility of the solution architect to take care of the domain knowledge challenge to decide which sensors will be used to sense what information that will be used later to solve a certain problem in certain field.

2nd challenge is the selection of the wireless communication and the application protocol used to transmit the data from the devices to IoT servers. These include selection from the various/multiple wireless communication technologies and application protocols available for use in a given geographical region.

3rd challenge is the planned storage size required to retain these accumulated data that will continuously keep floating from its numerous devices (huge number of devices) that are continuously sending data and will keep sending the data in the future. This means the storage size will have to be growing as well. This requires the ability to increase the storage without impacting the service. You cannot simply shutdown the service to change the disks in a server. It has to be an elastic storage to increase the disk space without disrupting the service offered to the devices.

4th challenge is the Realtime action required to be taken against all the huge amount of data coming from all of the huge amount of devices. This is known to be the most complex technical challenge that needs to be taken care of.

5th challenge is presenting the data to end users who would be using various types of devices with various capabilities. Users with IoT devices will be having computers, laptops, tablets, phones, watches and many more devices with limited display capabilities or even no display at all (i.e. few color LEDs or sound beeper).

Technical mandates

Realtime: Actions has to be in real time. Think of an IoT heartbeat pacemaker, there is no way to wait for some time before the device starts regulating the pace of patient heartbeat.

Availability: services offered by servers serving IoT solutions has to be highly available, no way it can go down. Again, think of a heartbeat pacemaker that reports patient status to a remote doctor in the hospital. There is no way these servers can go down.

Reliability: same scenarios result in the same results. If you have money in your prepaid phone account, you should be able to press dial and hear the ring tone on the other size.

Scalability: The more devices introduced in the network, the more the servers required to expand to respond to the growing traffic. If these servers are not automatically scaling, it should be easy to scale it up and down in few minutes if not seconds.

Project Management mandates.

Delivery lead times: Most of IoT projects have tight delivery schedule. That is the norm in IoT projects. Customers are going for IoT project implementation because there is a real and urgent need for an IoT solutions.

Rare competent resources: very few of the programmers and IT resources understands the details of IoT solutions and its challenges and ways of working. IoT solutions are not the standard front end and backend web development. It is much more than that.

Shortage in planned budget: it is usual to see IoT projects exceeding the planned budget and hence suffering from shortage in budget because customers would wish to have max functionalities and features in an IoT solution where in innovative IoT solutions, the skey is the limit and too many customer personnel from various teams and departments are involved in any IoT solution that means an IoT solution is required to full fill and satisfy the needs of everybody in the organization making it exceed the budget and time plan. It is always good to innovate and do new services but accommodating too many innovations during the project can lead to an undefined/unlimited scope and hence a failure to deliver the project.

MasterOfThings IoT AEP (Application Enablement Platforms) and similar IoT AEPs have been invented to solve challenges number 2, 3, 4 mainly and help in solving challenge number 5 as well as to cater for the technical mandates in addition to the project management mandates however the domain knowledge challenge is always out of scope of IoT AEP.

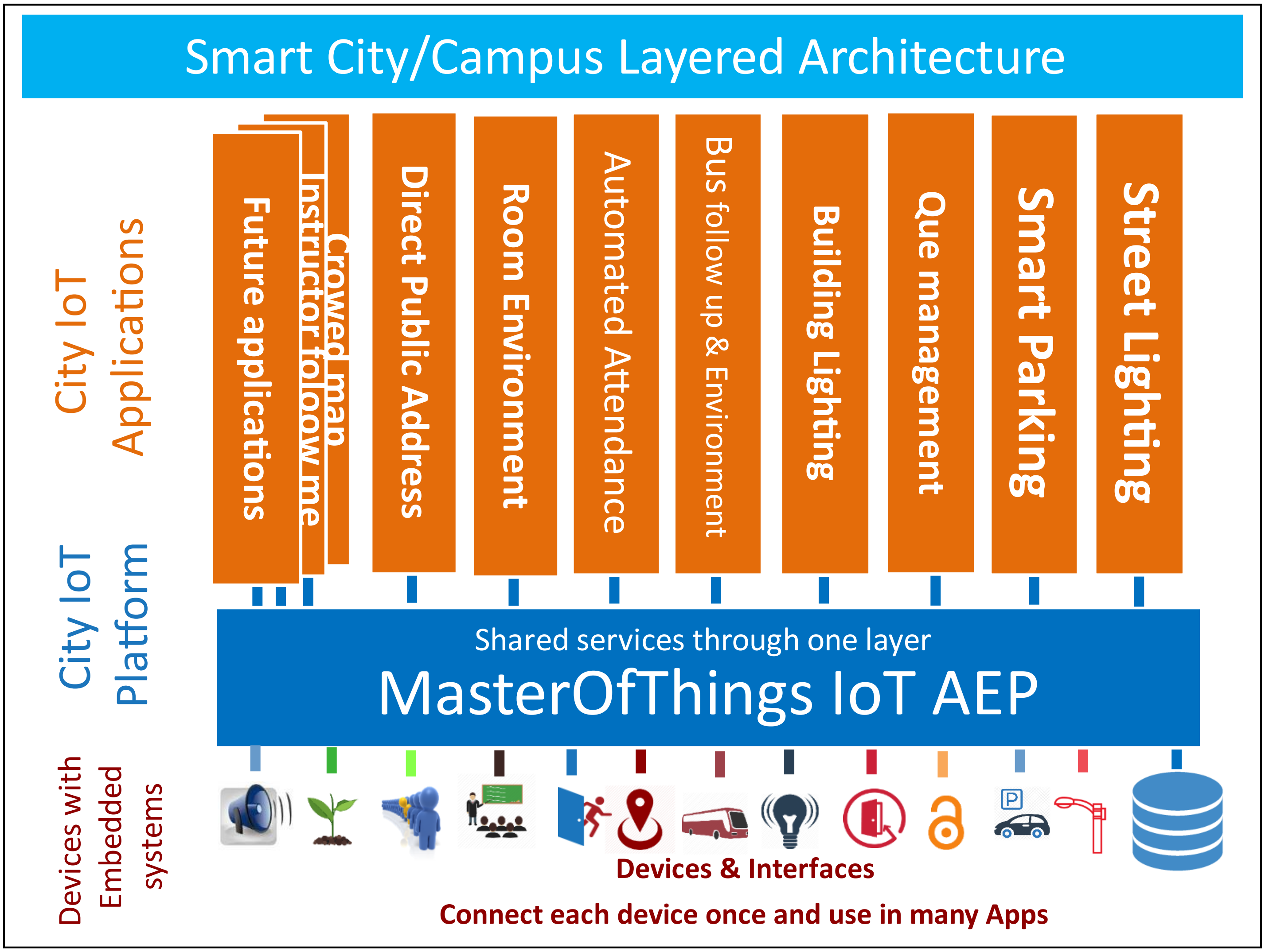
IoT applications development is totally different.

Some Bus builders think they can build a train buy connecting multiple buses ignoring the fact of speed, reliability, scalability and performance. **Similarly**, some organizations do not properly understand the IoT challenges , technical mandates and project management mandates assume IoT solutions and applications development are not different from traditional web development and some are even naive to think they can build professional IoT solutions relying on classical know-how or old technologies in GIS technologies or PHP or scripting languages 😊 while industrial automation engineers assume that SCADA (Supervisory Control And Data Acquisition) are suitable for building IIoT (Industrial IoT) solutions.

IoT AEP (Application Enablement Platforms) has been invented to solve IoT challenges number 2, 3, 4,5 and cater for the technical mandates as well as the project management mandate however the domain knowledge challenge is still out of scope of IoT AEP.

IoT AEP is simply put to make the development of the IoT solution in a much standard way both from technical perspective (like authenticating users, generating and managing alarms, etc.) and from an operational perspective (where all applications will be supported and operated using a standard operating procedures) that increase the productivity of the software developers and hence make more solutions available in less time and this makes the customers maximize the ROI of the project budget to in favor of features and functions in a solution that can be possible within the available project deliver time frame. Having said that, customer can get the solutions quickly and in an agile operation however still miracles cannot happen to bring every possible feature and function within few weeks 😊 specially if there is a need for a custom-made device and that happens a lot as well due to the need to innovative use cases where device makers did not build a ready-made device for every use case.

Layered architecture of Smart City, campus & gated community.



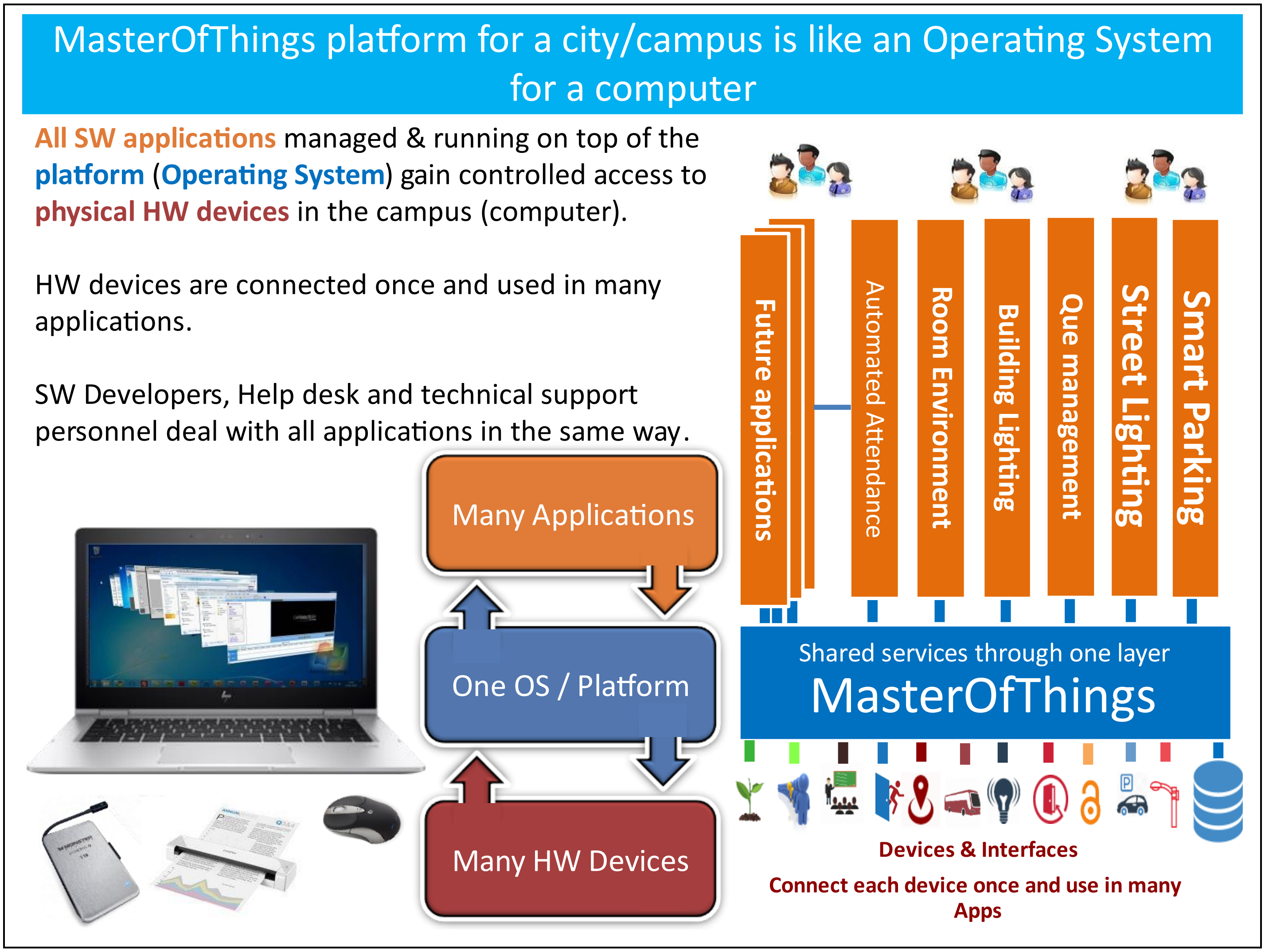
From an application developer perspective, without thinking of the communication layer, there would be 3 layers in any IoT solution.

The middle layer is the IoT platform that offer services to the above application layer and to the bottom layer of devices.

All the data are stored in the platform database (both memory based and disk based) whether it is application data, user data or devices data.

It is important to notice that once any device get connected to the platform, it can be utilized by many applications running on top the platform depending on settings and configurations defined by the platform administrator.

The indispensable role of a smart city IoT AEP.



Understanding the role of Smart City IoT Application Enablement Platform.

The IoT platform for a smart city is like the operating system for a computer.

We can easily understand the vital role of the smart city IoT AEP (Application Enablement Platform) in a smart city setup by comparing it to the role of the operating system for a computer.

- Common GUI and functionality in all applications and hence whoever can support word can also support power point and similarly in a city, who can support street lighting application can also support parking application, support waste application and so on.

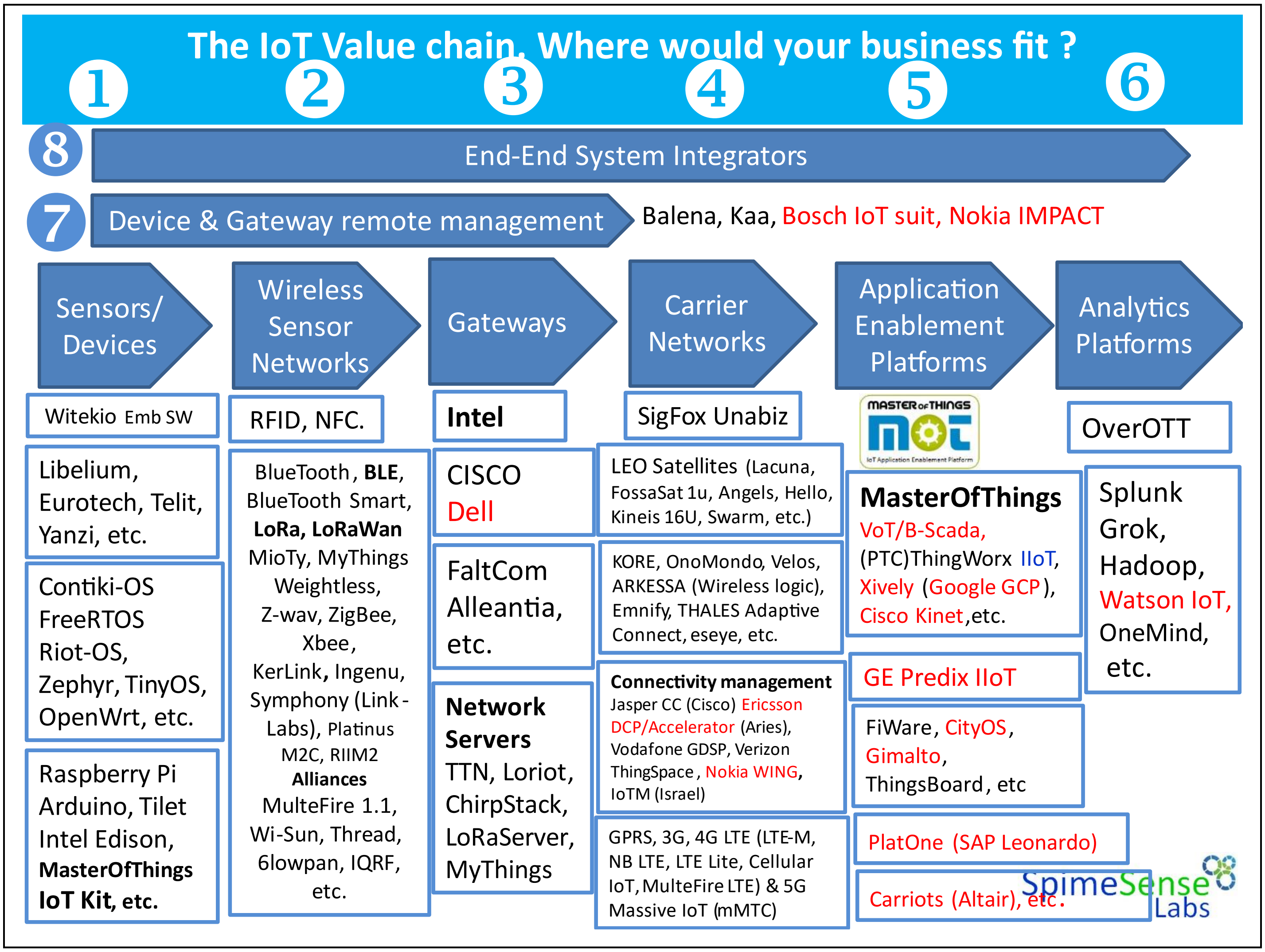
- Optimal utilization of Support resources, help desk and operations management teams (alarms versus events)
- Application management (performance, CPU utilization, disk utilization, network traffic, etc.)
- **Users Authentication and access rights management:** In a computer, the operating system is responsible for authenticating the users and managing their access and controls which user has the right to run which application. **Similarly in the city**, the platform is responsible for authenticating operating users and managing their access rights by categorizing users into groups and the platform controls which group of users are allowed to open which application.
- **Users Access Rights management:** In a computer, the operating system manages the access rights of the users not just towards the applications but also towards the devices. For example, the mouse is public common device that any user can use in any application however there could be multiple printers to the computer and each printer is dedicated to certain group of users or allowed to be used by certain group of users. **Similarly in the city,** a group of users are allowed to control and manage the lighting poles, its lighting schedule and lighting intensity while another group of users are allowed to control and manage the irrigation devices. Another group can manage and operate the traffic lights and so on.
- **Any device type:** In a computer, the operating system is responsible to recognize and communicate with all the attached devices (i.e. mouse, printer, etc.) **through a certain protocol implemented** in all the devices compatible with this operating system. **Similarly in the city**, the IoT Application Enablement Platform is responsible to recognize and communicate with all the attached devices (i.e. parking sensor, smart pole, waste bin fill level sensor, etc.) through a standard and widely used protocol (MQTT) that should be implemented in all the devices. Why MQTT specifically? that is a separate topic by its own but MQTT is the optimal choice simply because it is light weight, bidirectional and an ISO standard for many years. It would be really strange to find a device that is categorized as an IoT device or an IoT Gateway without MQTT support and too many question marks would arise and most probably this device won't fit the purpose of a smart city.
- **Device independency/freedom or Any supplier:** In a computer, the operating system empowers the computer owner to change any of the devices (i.e. the mouse) and bring an alternative compatible device from any brand without any impact on the applications on top of the operating system. **Similarly in the city**, the platform enables the city management to change any of its devices and bring an alternative compatible device from any supplier with minimal impact or no impact at all on the application running on

top of the platform. The device is considered as a compatible device as long as the device supports MQTT protocol.

- **Device authentication:** in a computer, once the user attaches a new device, the operating system notifies the user with basic information about the new attached device **however in a smart city, it is more important** that the IoT Application Enablement Platform authenticates all the devices and determine if this device should be connected to the city network and hence prevent the device from sending or receiving any data.
- **Devices data traffic encryption:** in a computer, the operating system sends data to (and receive data from) the attached devices without the need to encrypt this data as the device is directedly attached to the computer in front of the user **while in the city the situation is more complicated,** the IoT Application Enablement Platform is also in charge of decrypting all the data received from any device and encrypting all the control commands or configuration parameters that is sent from the platform to the device if the device is able to encrypt/decrypt communicated data and this encryption requires an encryption key that will differ from a device to another. That is a major role of the IoT Platform in the city to keep data about all the devices and encryption key for each device. It is important to note that not all of the devices in the city will have encryption keys. Some of the devices are not able to do encryption due its limited/constrained capabilities because of its limited functionalities and these devices should still be supported from an IoT Planform point of view.
- **Support for non-MQTT devices:** in a computer It is important to notice that the devices are made compatible with an operating system by the device manufacture and usually the device is branded as a device compatible with specific operating system or a set of operating systems (i.e. Windows device, mac device or android device). It is not the responsibility of the operating system to provide support for a certain device by a certain device manufacturer. **Similarly in a city**, the devices must support MQTT protocol to be compatible with the IoT application enablement platform used in the city. **It is a big architecture mistake to push the IoT Platform to support multiple protocols** for the sake of non-standard MQTT devices because the city will end up with a mix of protocols used in the city and that leads to too many issues in device authentication, data encryption, security, support and operation management and all of it becomes very complex to maintain and the city becomes a spaghetti city and will have to be redesigned to survive in the future.
- **Applications Management:** In a computer, the applications run on top of the operating system and under supervision of the operating system. **Similarly in the city,** all applications are running on top of the platform and under its supervision.

- **Applications data sharing:** in a computer, users copy data from one application to the other through the clip board. You can simply copy data from MS Word and past it to Power Point or copy data from MS Excel and past it in MS Word. It is the operating system responsibility to transfer/share these data from one application to the other. **Similarly in a city**, it is the responsibility of the City IoT AEP to share the data between applications and this requires the administrator to allow this data sharing between specific/certain applications under the overall data sharing policy.
- **Maturity of developing new applications:** in a computer, all the good operating systems provide a way to enable 3rd party developers, outside the operating system provider, to develop new applications that will run on top of the operating system. Actually the more successful the operating the system, the easier the tools that are used to develop new applications and this is why Microsoft provided GW Basic since old DOS (Disk Operating System) and provided such tools like visual basic since early windows days and now after many years of maturity Microsoft is offering visual studio for developers to build applications not provided by Microsoft and still these applications make use of all of the capabilities available in Microsoft windows operating system**. Similarly in the smart city**, the smart city IoT platform must come with tools to enable 3rd party developers to develop new applications. While a basic IoT platform will provide a set of APIs to application developers, a good IoT platform will provide SDK (APIs and Libraries) but an excellent (and mature) IoT Application Enablement platform will provide a visual IDE for 3rd party developers to quickly build/develop new applications
- **Scheduler or Timer:** to run certain tasks regularly in certain times or periodically every period of time.
- **Events, triggers or actions:** Rule based engine that triggers/fires certain actions/events when certain condition occur.
- **Common alarm viewer:** to view all alarms generated from all the applications. Something like MS Windows event viewer where you can find all the events that happened in all windows applications in one place.

Deep dive inside each building block of the IoT value chain



Let's dive in more details of each component (building block) of the value chain and the alternatives for each block and know the value of each block in the value chain and which company is doing what in this value chain.

The value chain consists of 8 components marked in the image from 1 to 8 and each of these components have sub-components grouped in categories with multiple alternatives for each group/category. The value of each of these alternatives is explained below with samples of the various alternatives in each sub-category.

Please note that I will continuously keep updated version of this illustration available in the LinkedIn article in the link https://www.linkedin.com/pulse/understanding-iot-value-chain-where-product-fits-bassem-boshra-hx5of/

1- Sensors/Devices:

These are the small computer devices usually called sensors and sometimes called objects that sense something to send it to the outside world.

Some device maker companies like Libelium build several IoT devices and have a large portfolio of devices. Other device makers are much smaller and most device makers can customize a device for a special use case. When a device maker starts creating a device, as a quick way to have something working to show it to the stockholder or customer, device makers usually start the device development/creation phase by utilizing a prototyping board like Arduino, Intel Edison or any of the commercially available hundreds of prototyping/development boards or even mini/small computers like Raspberry Pi.

If that prototype is good, the device maker through it all away and start building the real actual/final device by choosing the right low power wireless technology modem chip, the right MCU chip, enough memory chip, screen, buttons, sensing elements/modules and other components required for the device and fix all of these components together on custom PCB (Printed Circuit Board) and package all of that inside a suitable enclosure/box.

These small devices have software that is running on top of an operating system inside the device. This device software is usually called device firmware or embedded system software. Some companies like Witekio are specialized in writing/developing the embedded device firmware only and not making the whole device.

Depending on the complexity of the functionality required in the device, the device maker will choose a suitable RTOS (Real Time Operating System) for their device.

Real Time Operating System:

Similar to a mobile phone, an IoT Devices need to have RTOS. There are lots of Operating systems whether of commercial license or open source. A very interesting open source RTOS list with a brief description of each RTOS and its supported HW platforms/architecture and supported wireless technology are available at OSRTOS.com

The selection of the suitable RTOS depends on the specific requirements of the required embedded system including device resource constraints, real-time capabilities, scalability needs, community support and the intended use case. Contiki-OS and Riot-OS are tailored for IoT devices, while FreeRTOS and Zephyr are more general-purpose, and TinyOS is specialized for wireless sensor networks. OpenWrt, on the other hand, is suitable for bigger devices used as networking equipment (IoT Gateway).

Zephyr and Riot-OS are particularly well-rounded for MQTT-enabled IoT applications, while the others, depending on the time of implementation, may require additional customization or external libraries to support MQTT protocol

Here is a brief comparison between some famous RTOS, thanks to ChatGPT and Gemini

Zephyr:

A. *Type:* Zephyr is a real-time operating system designed for resource-constrained devices, with a focus on scalability and security.

B. *Features:* Supports various architectures, has a modular design, and offers comprehensive documentation.

C. **MQTT Support:** Zephyr includes MQTT protocol support, making it suitable for IoT applications.

D. Scalability: Supports a range of devices, from resource-constrained to more powerful systems.

E. Community Support: Zephyr has a robust community and is backed by the Linux Foundation, ensuring ongoing development and support.

Riot-OS:

A. *Type:* Riot-OS is an open-source microkernel-based operating system designed for IoT devices and low-power embedded systems.

B. *Features:* Supports a wide range of devices, low memory footprint, and has a modular architecture.

C. **MQTT Support:** Riot-OS includes a lightweight MQTT implementation suitable for IoT devices.

D. Scalability: Designed for resource-constrained devices, making it suitable for IoT and embedded systems.

E. Community Support: Has a growing community with increasing support for various IoT protocols.

Contiki-OS:

A. *Type:* Contiki-OS is an open-source, highly portable, and lightweight operating system designed for the Internet of Things (IoT) devices.

B. *Features:* Supports IPv6 networking, low-power consumption, and is known for its efficiency in resource-constrained devices.

C. **MQTT Support:** Contiki-OS provides MQTT support through its Contiki **MQTT-SN** (Sensor Network) implementation.

D. Scalability: Suitable for resource-constrained devices and low-power IoT applications.

E. Community Support: Has an active community, but may have fewer resources compared to larger projects.

FreeRTOS:

A. *Type:* FreeRTOS is an open-source real-time operating system (RTOS) designed for embedded systems.

B. *Features:* Provides a preemptive scheduling kernel, small memory footprint, and is widely adopted in various industries.

C. **MQTT Support:** FreeRTOS supports MQTT through **external libraries** and projects that can be integrated for MQTT functionality.

D. Scalability: Primarily designed for resource-constrained devices, suitable for microcontrollers and small embedded systems.

E. Community Support: Has a large and active community, which can be beneficial for finding relevant third-party libraries.

TinyOS:

A. *Type:* TinyOS is an open-source, event-driven operating system designed for wireless sensor networks (WSN).

B. *Features:* Optimized for low-power and memory-constrained devices, supports component-based programming, and is well-suited for WSN applications.

C. **MQTT Support:** TinyOS **does not** have native MQTT support. Integration of **external libraries** or development may be required.

D. Scalability: Designed for low-power wireless sensor nodes, tailored for sensor networks.

E. Community Support: Has an established community, but its focus on specific applications might limit general-purpose support.

OpenWrt:

A. *Type:* OpenWrt is a Linux distribution for embedded devices **suitable for** gateways, routers and networking equipment.

B. *Features:* Provides a customizable Linux environment for routers, supports package management, and allows users to install additional software.

C. **MQTT Support:** OpenWrt supports MQTT through **additional packages** that can be installed.

D. Scalability: Initially developed for routers, OpenWrt is suitable for a range of devices including IoT gateways.

E. Community Support: Boasts a large and active community, with extensive documentation and package availability.

Conclusion:

- **Open Source:** All listed RTOS (Contiki-OS, Riot-OS ,Zephyr, TinyOS and OpenWrt) except FreeRTOS are fully open source, offering better customization and community support.
- **MQTT:** All above RTOS support MQTT directly or through 3rd party libraries, making them suitable for IoT devices.
- **Ease of Use:** ContikiOS and TinyOS are easiest for beginners, while Zephyr and OpenWRT have steeper learning curves.
- **Project Suitability:** Choose based on project requirements:
    - **Resource-constrained:** Zephyr, RiotOS, ContikiOS
    - **Low-power:** ContikiOS, TinyOS
    - **Real-time:** FreeRTOS
    - **Networking:** OpenWRT
- **Main Features:** Consider Zephyr for security and multi-core, RiotOS for low power, ContikiOS for IPv6, and OpenWRT for extensive network features.

2- Wireless Sensor Networks:

Wireless communication to transfer the data through small distance. This distance can be very small up to 1 cm like the case in NFC and can be few meters (up to 10-15 m) in case of RFID, larger distance like BLE (up to 300 m in BLE 5) or few kilometers like the case of LoRa. All wireless sensor networks are meant to be for low power consumption hence the name LPWAN (Low Power Wide Area Network) and the wireless technology save the device power either by minimizing the data throughput or minimizing the distance to save the device battery.

Experts' advice on What is the best way to design an IOT network for a smart city, is available on the link https://www.linkedin.com/advice/1/what-best-way-design-iot-network-smart-city-ynb8c?utm_source=share&utm_medium=member_android&utm_campaign=share_via

3- Gateways:

The gateway is supposed to take the data from a small distance wireless technology (i.e. BLE, LoRa, etc.) and transfer it to a long-distance carrier network (sometimes up to hundreds of kilometers). The better the gateway the more wireless technologies it can support and that is why companies like intel develop a board with modem chips for multiple wireless technologies and supply it to the gateway makers (i.e. Cisco) to use it inside their gateways with intel inside logo.

some companies like Faltcom started building industrial gateways that can accommodate working in tough conditions. later it has been acquired by telia and now branded as Telia IoT Edge that is used as a gateway to connect public buses in Sweden to the internet.

Alleantia also built an industrial IoT gateway that is now branded as industrial edge platform.

Please note that Gateways represent the edge of the network and hence edge applications run on the gateways. A detailed explanation of the edge and why we need edge applications is available in the link below

https://www.linkedin.com/posts/bassem-boshra-9216271_mahara-tech-all-courses-activity-7134140573141061632-gr03?utm_source=share&utm_medium=member_desktop

large networks contain multiple gateways to cover larger areas. Each of the gateways needs to authenticate the device and check if the gateway will allow the device to use the gateway to connect to network or not. Each gateway cannot decide this by itself but rather rely on a higher decision authority known as the network server that take a central decision for all the gateways in the same network when it comes to each device.

LPWAN Connectivity Management Platforms (Network Servers):

A. **The Things Network (TTN):**

1. TTN is a community-driven initiative that provides an open-source LoRaWAN network server. It allows users to set up their own private network or join the global public community network. The source code is available on GitHub.

B. **ChirpStack:**

1. Formerly known as LoRa Server, ChirpStack is an open-source LoRaWAN network server that provides scalability and flexibility. It is written in the Go programming language and is available on GitHub.

C. **LoRaServer:**

1. LoRaServer is an open-source LoRaWAN network server developed in Go. It supports the LoRaWAN 1.0 and 1.1 specifications and provides a RESTful API for integration. The source code is accessible on GitHub.

D. **OpenChirp:**

1. OpenChirp is an open-source project that includes a LoRaWAN network server. It is designed to be scalable and extensible, allowing for customization. The project is available on GitHub.

E. **Loriot:**

1. Loriot offers an open-source version of its LoRaWAN network server. It supports various deployment scenarios, including public, private, and hybrid setups. The source code can be found on their GitHub repository.

F. **MyThings:** MyThings is developed by Behrtech. MyThings is a MioTy central network connectivity platform with device management capabilities designed for MIOTY-based low-power wide-area networks. In this context, MyThings is comparable to Network Server and Application server in LoRaWan.

4- Carrier Networks:

Carrier networks are networks that can carry the data to long distances in several km distance. These networks could be mobile network (2G, 3G, 4G, 5G) operated by mobile CSP (Carrier Service Provider) Like Orange, Vodafone, Dutch Telecom in Germany or Verizon in USA. The carrier network could be copper cables or fiber cables operated by fixed carrier operator known as fixed CSP like WE (Telecom Egypt) or a carrier network could be a LEO (Low Earth Orbit) satellite network operated by satellite network provider like Lacuna, Swarm/Star Link or one web.

A. **SigFox started the first low power carrier operator** business by catering for saving device power over data throughput allowing devices to service for 7 years on a battery without recharge (actually replacement) and had their networks in 70+ countries as one global network that is acquired later by UnbaBiz and became SigFox UnaBiz. Because 3G and 4G consume a lot of power from the device battery, the competition from Sigfox pushed the network makers/vendors to look into ways to make their networks consume less power from the device and giving away the high data throughput. Ericsson proposed

NB-IoT/NB LTE, Nokia proposed LTE light and Huawei proposed cellular IoT and at the end **3GPP standardization body adapted Ericsson proposed changes in the 4G network to enable mobile operators/CSP started offering NB-IoT/NB LTE for IoT devices to have low power mobile communication** but they still inherit the problems in roaming agreements if the device will be moving between countries where data rates are higher in the visiting countries compared to the device home network in its home country. 5G networks are coming by default with NB-IoT (mMTC) mode of operation for IoT devices.

B. **Connectivity Management Platforms:** Mobile operators figured out that they would need to have an IoT CMP (Connectivity Management Platform). CMP is a software that make it easier to manage several connectivity subscriptions (Sim cards) for a specific customer (usually a corporate) where the customer would need to enable/disable the subscription or monitor the amount of data generated from every SIM (representing a device). The first company that build a connectivity management platform was Jasper wireless that have been acquired later by Cisco and later the product is branded Cisco Connect and now later branded as Cisco Control Center. Mobile network providers/vendors (like Ericsson, Nokia Huawei) also started building their own CMPs. Ericsson made Ericsson DCP with some billing capabilities as well. later Ericsson branded it as Ericsson Accelerator and later sold it to Aeris. In 2017 Nokia started building its Nokia WING (Worldwide IoT Network Grid) as its CMP and some operators also built their own CMP like Vodafone DCP later called it Vodafone IoT and in 2024 Vodafone moved this business out of Vodafone in a new separate joint venture company between Microsoft and Vodafone. Similarly, Verizon in USA build their own CMP and branded it ThingSpace.

C. **To solve the roaming problem in NB-IoT**, some companies like KORE, Arkessa (acquired by wireless logic) started utilizing eSIM capabilities to provide a global sim to eliminate the high roaming data tariffs for traveling devices for the global device makers who would like to monitor their devices wherever these devices would in the world. Other companies followed this model include Emnify, Onomondo, velos, eseye, etc.

5- Application Enablement Platforms:

An IoT AEP is the smartest and most complex component in the IoT value chain where all IoT the magic happens. Few people (non-field experts) refer to it as the IoT middle ware because it is a middle layer between devices and applications.

There are a lot of companies claim that they offer an IoT AEP. **An IoT AEP must offer certain capabilities and have certain features to be considered application enablement platform. These are discussed in details in a separate dedicated slide/section/article. However, in this section we will divide the AEP into families.**

A family that focuses on boosting the productivity and efficiency of IoT application development process.

This family includes MasterOfThings IoT and PTC ThingWorx IIoT (Industrial IoT). Platforms in this category are all licensed software whose main purpose is to empower the developers to do more applications quickly, efficiently and properly usually by providing 4 main core modules of the IoT Platform and those modules are visual IDE (integrated development environment) to faster the development of the application, sensor profile definition to define device sensing capabilities, data collection and acquisition module to collect data from the devices and an Administration management interface to manage end users and developers access rights towards application and devices.

An open-source family that offers free community version and an enterprise licensed version.

This category focus more on providing a mix of libraries for free use by the developers to use to develop their own applications. **The best in this category is ThingsBoard** and it is developed by a company in Ukraine. **The most complex in this category is FiWare** as an abbreviation for Future Internet Ware and it is developed as financed R&D contributions from various European interties including companies, research centers and universities all financed to add enhancements or features as much as they like to add to FiWare and that made FiWare a complex mix of libraries for whatever you might need to develop future undefined software.

Category owned by SCADA (Supervisory Control and Data Acquisition) solution providers.

Where they all focus on IIoT (Industrial IoT) that will replace SCADA systems. **Almost all successful and the big Scada providers decided to acquire an IoT platform** where PTC acquired ThingWorx, Schneider acquired WonderWare, and Siemens acquired MindSphere. On the contrary GE (General Electric) had a well-known failure story because they decided to build their own Predix IoT Platform by their own with no knowledge or experience in IoT and assuming it is similar to Scada.

Data base engine makers category

Data base engine makers (like SAP) acquired software that can collect data from the devices and store it inside the database engine and that allows them to use their existing development tools and community of partners to develop applications based on the data stored in the database as they used to do. An example of this category is SAP that acquired PlatOne and branded it as SAP Leonardo.

Messaging bus category

platforms that tried to implement the messaging bus technology used in communication between various systems inside complex IT environments (i.e. the IT environment in mobile telecom operators and banking institutions) as a messaging bus for devices to communicate to each other and communicate to surrounding servers. Carriots was driving this category and later acquired by Altair and it represents all the failing IoT Platforms in this category.

6- Analytics platforms:

These are platforms used to analyze all the big data collected over many years from various IoT applications running in a city or a large corporate/enterprise to figure our new relation between the collected data sets. Hadoop is the most well-known open-source big data analytics platforms and the rest are licensed commercial software.

7- Device & gateway remote management platforms:

These are platforms used to remotely configure the devices and send remote software updates. Similar to the Play store functionality in mobile phones but more for devices. Kaa is an open-source device management platform while the rest are licensed commercial software.

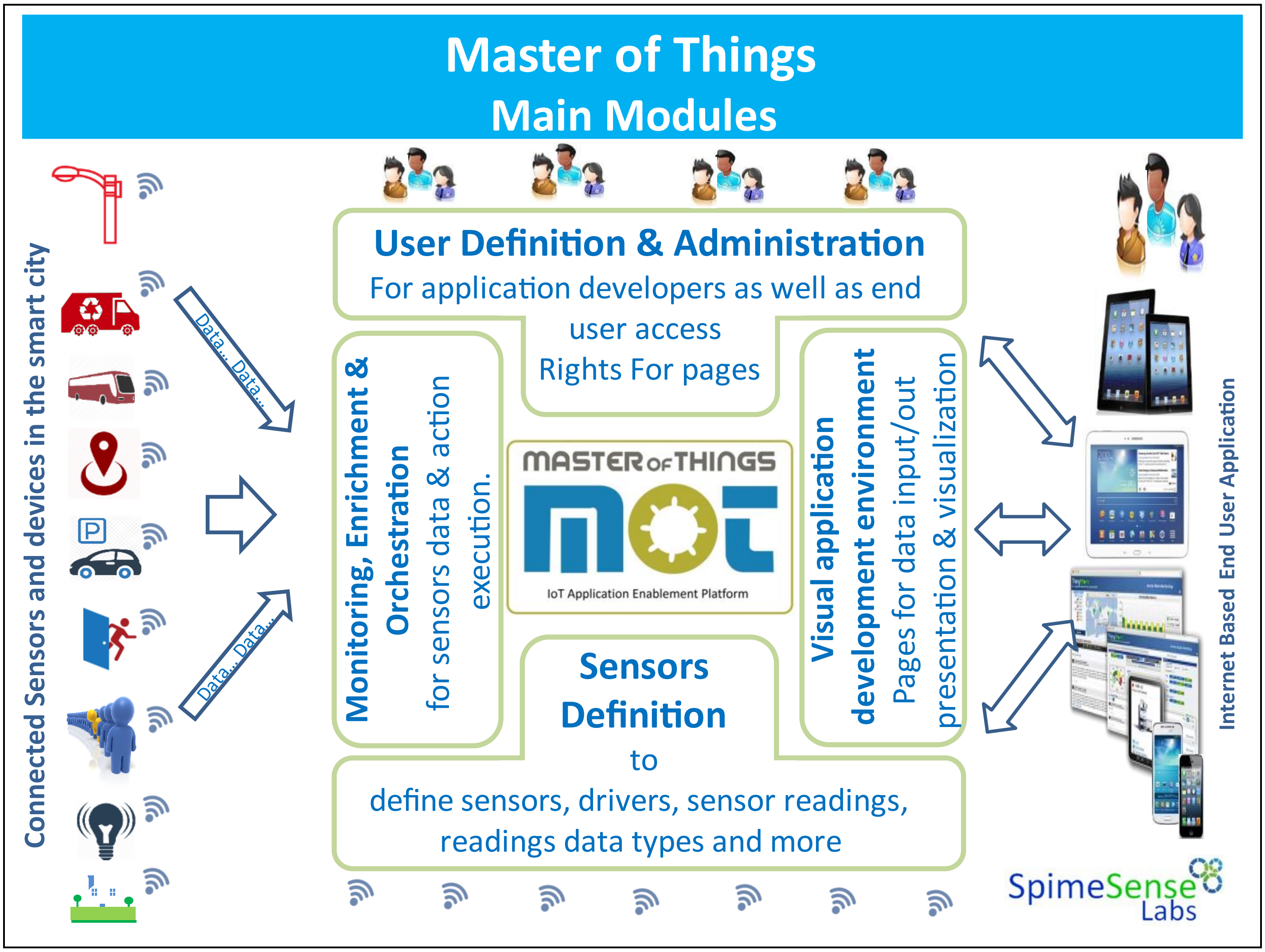
8- End-to-End system integration companies

These are companies that have competent resources who have skills and knowledge in every component of the IoT Value chain. Those are the companies contracted by the customers to develop and complete end to end solution utilizing a mix of the IoT value chain components. The most important skill that must exist in these companies is the solution architect role. Then the application developers and system integrators and the project managers.

Details of the job roles and skills required for each role are explained in the free online career talk 10 minutes video available on MaharaTech learning portal in the link below.

https://maharatech.gov.eg/course/view.php?id=1325

The main modules of true IoT Application Enablement Platform



A proper IoT Application Enablement Platform must have 4 main modules

The IoT AEP is a platform to enable both rapid, proper and sustainable development, operation and management of IoT applications. IoT platforms are sometimes referred to as middleware solutions, which offer a set of pre-packaged functionalities to help the developer to build IoT application quickly. The pre-packed functionalities required for IoT application development includes various data collection protocols (i.e. **MQTT**, CoAp, HTTP, etc.), data encryption, data storage (database engine), on the fly data investigation, triggering events if the data matches a specific pattern defined by the developer, user access rights management functionality, data visualization components (charts, graphs, digital display, analog meter pointers etc.).

The aim of an IoT AEP (Application Enablement Platform) is to reduce the time, effort and cost of getting new IoT solutions **properly** built, implemented, operated and supported. There are several layers to an IoT solution and these are becoming increasingly complex as the market develops. The IoT platform takes advantage of the fact that the majority of what is needed in IoT solutions is the same and does not need to be redeveloped for every application. Almost 80% of IoT solutions are made up of common parts and made available through an IoT platform. The platform then also provides the means for customizing and configuring the solution for a specific application need.

IoT AEP is neither an application nor a mean of managing assets (devices). **AEP is meant to provide all the foundational services** that are required to build a proper IoT solution from the ground up. Foundational service includes application communication protocols; data storage; management of users; application building and enablement; user interface development; security mechanisms; real time analytics, data valuations and visualization.

IoT AEP is sometimes referred to as IoT middle ware because AEPs are a fundamental building block and interface with almost every component in the system, including enterprise backends, IoT devices and supplementary services. It is important to highlight that AEP is neither a framework nor a group of software libraries and it is not a group of applications suites. It is not a software development stack.

A proper IoT Application Enablement Platform must have 4 main modules (listed below) that can't be missed by any mean. Each of those modules would include multiple functionalities to fulfill the module capabilities. Below are the 4 main modules of an IoT AEP however the detailed features (feature by feature description) of each module is out of scope of this document.

A. **Sensors & Devices Profile Definition:**

   The platform should provide a mean to define sensors, drivers, sensors readings, reading data types, data acquisition application protocols used by the sensors and their gateways, etc.

B. **Monitoring, Enrichment, Orchestration and action execution:**

   The platform must provide a mean to monitor sensor data on real time, enrich the sensors data, provide a rule-based engine to orchestrate the sensor date flow between systems and instantly executes defined actions when these data match some related criteria.

C. **Users access rights management and administration:**

   The platform should provide means to manage users (both the application developers and end users) access rights towards all the IoT application components/pages. For developers, this module of the of the IoT AEP manages the access right to create the application, update/delete parts of the application, publish it to end users. For end users, this module of the of the IoT AEP manages the access right to access/open the application and execute the application.

D. Visual Integrated Development Environment (IDE):

The IDE is considered as the most important functionality of the platform for a developer. The platform should provide a visual development environment for developers to use to develop and build the logic of various applications (i.e. web application and/or mobile applications) for data input/output presentation and visualization or at least to build dynamic dashboards.

The visual IDE enables developers to write both back-end logic and front-end visualizations. Being a visual development environment, will boost the developer productivity and make it possible for developers to show the customer the application while it is being developed, hence, get instant feedback from the customer.

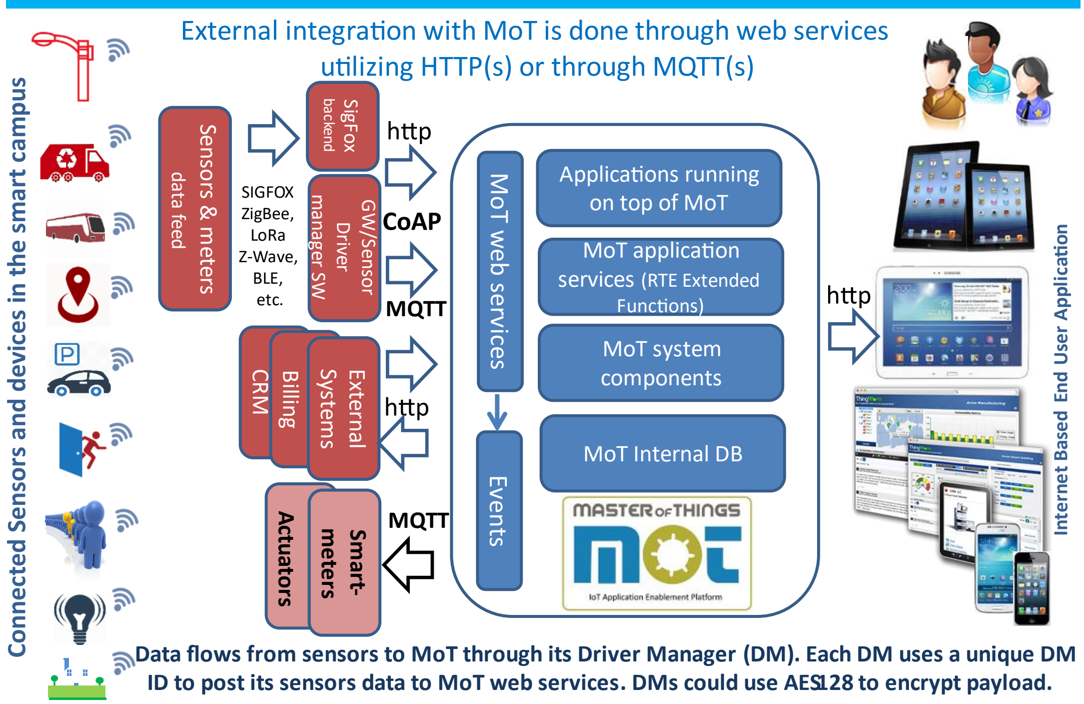
Master of Things

Comes with optional Applications for smooth IoT & M2M operations

MoT offers some out of the box optional Applications usually needed/requested by enterprises to run IoT/M2M applications. These applications are built by MoT IDE and its code is provided to solution providers to customize it according to their customer needs.

- **Alarm management :** map based application to manage alarms generated for sites/locations based on sensors data (missing , invalid, out of range, etc.)
- **Task management :** application to create & manage tasks towards organization workforce (individuals, teams/groups/departments, etc.) to check a device, check connectivity, power, etc..
- **Asset management:** map based application to manage sites fixed assists and sensors (installation, commissioning, movement, decommissioning, etc.)
- **Global Calendar:** application to manage (meetings, events, vacations, etc.) towards organization workforce teams/groups/departments.

MasterOfThings Layers architecture, Integration and Data flow



Data flows from sensors to MoT through its Driver Manager (DM). Each DM uses a unique DM ID to post its sensors data to MoT web services. DMs could use AES128 to encrypt payload.

Thank You

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